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Advanced electronic structure calculation of semiconductors

PHYSICS PROJECT II

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Abstract

This report develops an efficient heat-capacity workflow for methane-loaded MOF-5 using a machine-learned interatomic potential (MLIP). Classical flexible-cell NPT molecular dynamics (MD) samples the loaded system and retains anharmonic methane and host-guest motion. Automatic differentiation (AD) through the matching PET-JAX model provides sparse Cartesian Hessians and normal-mode frequencies at quenched reference structures, which must be validated as local minima. Rather than reproduce the reference simulations with an MLIP, the workflow applies the hybrid approximation proposed by Kapil *et al.*: the loaded-system classical anharmonic heat capacity is corrected by the difference between quantum and classical harmonic heat capacities evaluated for the same loaded composition and PET potential. The already-equilibrated empty MOF-5 structure is relaxed directly and evaluated with one reference Hessian; no empty-framework MD campaign is required. The resulting low-temperature and model-dependent accuracy must be validated for the selected MLIP. Finite-trajectory sampling error and correlated classical-harmonic model spread are evaluated with a last-layer prediction-rigidity (LLPR) ensemble. The available PET-MAD and PET-SOL calculations cover 50, 100, and 150 methane molecules and give exploratory room-temperature hybrid values near 0.93, 1.07–1.08, and 1.16 J/g/K, respectively. Imaginary Hessian modes, incomplete relaxation, production drift, and strong sensitivity to uncertainty propagation prevent interpreting these values as converged predictions. Variation across independently quenched minima remains unestimated.

1 Scientific motivation and scope

Metal-organic frameworks (MOFs) combine inorganic nodes with organic linkers to form porous crystalline networks. Their vibrational spectrum contains high-frequency local bond vibrations as well as soft linker, pore, and collective framework modes. Because the contribution of a quantum vibrational mode depends strongly on the ratio of its energy to $k_B T$, a broad frequency distribution produces a strongly temperature-dependent heat capacity. Guest adsorption adds translational, rotational, and host-guest degrees of freedom that may be weakly confined and strongly anharmonic.

The material considered here is MOF-5. Its conventional cubic cell contains 424 framework atoms with composition $C_{192}H_{96}O_{104}Zn_{32}$. In the loaded calculations, 50, 100, and 150 methane molecules give 674, 924, and 1174 atoms, respectively. The central 100-molecule example has composition $C_{292}H_{496}O_{104}Zn_{32}$. The principal objectives are:

1. to sample the anharmonic thermodynamics of methane-loaded MOF-5 with classical NPT MD and independent replicas;
2. to compute force constants efficiently by sparse

AD at optimized empty and loaded structures;

3. to obtain and converge the loaded-system harmonic quantum correction;
4. to combine that correction with the classical enthalpy derivative to estimate the loaded-system C_P efficiently; and
5. to establish a reproducible route from short cluster validation jobs to statistically meaningful production calculations.

The workflow has three computational stages. Loaded-system MD supplies the classical anharmonic enthalpy curve. Fixed-cell optimizations followed by AD Hessians supply quantum-minus-classical harmonic corrections. A final analysis combines the two terms with consistent composition and mass normalization. The empty-framework Hessian is retained as a reference and validation calculation, not as a replacement for the loaded-system correction.

The aim is to apply and assess this hybrid approximation on the PET potential, not to reproduce the numerical curves, force field, methane-insertion ensemble, or simulation protocol of Kapil *et al.* Agreement with a reported reference value can be a useful external comparison, but is not the criterion for success. The criteria are a well-defined, consistent PET potential across all terms, converged sampling and Hessians, and uncertainty estimates for the resulting hybrid curve.

2 Theory

2.1 Classical molecular dynamics

Molecular dynamics (MD) connects an interatomic potential to observable properties by integrating equations of motion and averaging over the resulting trajectory. A calculation therefore requires both a mechanical model, which specifies the forces, and a statistical ensemble, which specifies the thermodynamic conditions and probabilities of microscopic states. The following development introduces the foundations needed for equilibrium, fixed-composition simulations and for the heat-capacity calculations in this report. Standard treatments are given by Allen and Tildesley and by Tuckerman [1, 2].

2.1.1 Microscopic model and equations of motion

Within the Born-Oppenheimer approximation, the electrons define a potential-energy surface for the nuclei. Classical MD makes the additional approximation that the nuclei are point particles obeying Newtonian mechanics; this does not follow from the Born-Oppenheimer approximation alone. Let N be the number of atoms, $\mathbf{R} = (\mathbf{R}_1, \dots, \mathbf{R}_N)$ their positions, and $\mathbf{p} = (\mathbf{p}_1, \dots, \mathbf{p}_N)$ their momenta, with $\mathbf{p}_i = m_i \mathbf{v}_i$. The cell matrix $\mathbf{h}$ contains the three lattice vectors as

columns, and $V = \det \mathbf{h} > 0$ is its volume. At fixed cell, the physical Hamiltonian is

$$\mathcal{H}(\mathbf{R}, \mathbf{p}; \mathbf{h}) = K(\mathbf{p}) + U(\mathbf{R}; \mathbf{h}), \quad K = \sum_{i=1}^N \frac{\mathbf{p}_i^2}{2m_i}. \quad (1)$$

Here U denotes potential energy, $E = \mathcal{H}$ denotes instantaneous total internal energy, and $H = E + P_{\text{ext}}V$ will denote physical enthalpy. The scalar pressure P_{ext} is distinct from the momenta $\mathbf{p}$. Chemical species are fixed and are suppressed in the arguments of U .

Hamilton's equations are equivalent to Newton's second law:

$$\dot{\mathbf{R}}_i = \frac{\partial \mathcal{H}}{\partial \mathbf{p}_i} = \frac{\mathbf{p}_i}{m_i}, \quad \dot{\mathbf{p}}_i = -\frac{\partial \mathcal{H}}{\partial \mathbf{R}_i} = \mathbf{F}_i. \quad (2)$$

Thus $\mathbf{F}_i = -\nabla_i U$ and $m_i \ddot{\mathbf{R}}_i = \mathbf{F}_i$. The potential may be an analytical force field, an electronic-structure calculation, or an MLIP; the statistical-mechanical framework is the same. A conservative, time-independent potential conserves E under exact fixed-cell dynamics. Translational invariance also conserves total momentum in the absence of external forces. Thermostats and barostats modify these physical equations to exchange energy and volume with modeled reservoirs.

Classical nuclear motion omits zero-point energy, tunneling, and quantum occupation statistics. For a vibration of angular frequency ω , the classical approximation is most appropriate when $k_B T \gg \hbar \omega$. High-frequency vibrations can violate this condition even when slower collective motions are approximately classical. This distinction motivates the separate harmonic quantum correction used later in the report.

2.1.2 Phase space, probability, and equilibrium averages

A microstate is a point $\Gamma = (\mathbf{R}, \mathbf{p})$ in phase space; a macrostate specifies quantities such as N , V , and T . An ensemble is a probability distribution over microstates compatible with those conditions. For unconstrained Cartesian coordinates, a conventional dimensionless measure is

$$d\Gamma = \frac{d^{3N}R d^{3N}p}{h_P^{3N} \prod_s N_s!}, \quad (3)$$

where h_P is Planck's constant and N_s counts identical particles of species s . The scalar h_P is unrelated to the cell matrix $\mathbf{h}$. These constant normalization factors cancel in fixed-composition averages; Planck's constant here fixes the phase-space measure and does not make the dynamics quantum mechanical. Constraints require the corresponding reduced phase-space measure. The explicit partition functions below first refer to the unconstrained system.

For a normalized equilibrium density $\rho(\Gamma)$, an observable $A(\Gamma)$ has expectation

$$\langle A \rangle = \int d\Gamma \rho(\Gamma) A(\Gamma), \quad \int d\Gamma \rho(\Gamma) = 1. \quad (4)$$

The phase-space density evolves according to Liouville's equation,

$$\frac{\partial \rho}{\partial t} + \sum_i \left(\dot{\mathbf{R}}_i \cdot \nabla_{\mathbf{R}_i} \rho + \dot{\mathbf{p}}_i \cdot \nabla_{\mathbf{p}_i} \rho \right) = 0. \quad (5)$$

Hamiltonian flow preserves phase-space volume, so the density is constant along a trajectory. This property explains why a density depending only on a conserved Hamiltonian is stationary, but does not establish that one trajectory explores the entire distribution.

Ergodicity is the assumption that a sufficiently long trajectory samples the accessible equilibrium distribution, so that

$$\langle A \rangle = \lim_{\tau \rightarrow \infty} \frac{1}{\tau} \int_0^\tau A(\Gamma(t)) dt. \quad (6)$$

A finite trajectory estimates this average only after equilibration and only if relevant states are visited. Barrier crossing, guest rearrangement, and slow framework deformation can be much slower than local vibrations. A stationary temperature alone is therefore insufficient evidence of adequate sampling. Independent replicas and structural diagnostics are needed to identify trapping and incomplete exploration.

2.1.3 Microcanonical, canonical, and isothermal-isobaric ensembles

An isolated system at fixed N , V , and total energy E_0 has the microcanonical (NVE) density

$$\rho_{NVE}(\Gamma) = \frac{\delta(E_0 - \mathcal{H}(\Gamma))}{\Omega(N, V, E_0)}, \quad (7)$$

where $\Omega = \int d\Gamma \delta(E_0 - \mathcal{H})$ is the density of states and δ is the Dirac delta distribution. This expresses equal probability on the accessible energy shell. Other exact invariants, such as a fixed total momentum, further restrict that shell. With a fixed reference energy interval δE , the entropy can be written $S = k_B \ln(\Omega \delta E)$, and temperature follows from $T^{-1} = (\partial S / \partial E_0)_{N, V}$. NVE dynamics is useful both for physical dynamical properties and for checking numerical energy conservation.

A system exchanging energy with a much larger heat bath instead has fixed N , V , and bath temperature T . Expanding the reservoir entropy to first order in the system energy gives the Boltzmann weight $\exp(-\beta \mathcal{H})$, where $\beta = (k_B T)^{-1}$. Consequently,

$$Z(N, V, T) = \int d\Gamma e^{-\beta \mathcal{H}(\Gamma)}, \quad (8)$$

$$\rho_{NVT}(\Gamma) = Z^{-1} e^{-\beta \mathcal{H}(\Gamma)}. \quad (9)$$

Z is the canonical partition function. Gaussian integration over momenta leaves a configurational density proportional to $e^{-\beta U(\mathbf{R})}$; positions and momenta factorize for this separable Hamiltonian. The Helmholtz free energy and thermodynamic derivatives are

$$F = -k_B T \ln Z, \quad \langle E \rangle = - \left(\frac{\partial \ln Z}{\partial \beta} \right)_{N, V}, \quad (10)$$

$$S = - \left(\frac{\partial F}{\partial T} \right)_{N, V}, \quad P = - \left(\frac{\partial F}{\partial V} \right)_{N, T}. \quad (11)$$

The same entropy is $S = -k_B \int \rho \ln \rho d\Gamma$. These relations connect microscopic state probabilities to macroscopic thermodynamics. Absolute free energies generally require additional methods, such as thermodynamic integration, because ordinary trajectory averages do not directly supply the normalization Z .

Allowing volume exchange with a pressure reservoir gives the isothermal–isobaric (NPT) ensemble. For a cell of fixed shape with variable volume, define

$$\Delta(N, P_{\text{ext}}, T) = \int_0^\infty \frac{dV}{V_0} e^{-\beta P_{\text{ext}} V} Z(N, V, T), \quad (12)$$

where V_0 is a fixed reference volume making Δ dimensionless. Relative to $d\Gamma dV/V_0$, the density is

$$\rho_{NPT}(\Gamma, V) = \Delta^{-1} e^{-\beta[\mathcal{H}(\Gamma; V) + P_{\text{ext}} V]}. \quad (13)$$

Thus $G = -k_B T \ln \Delta$ is the Gibbs free energy in this convention, $\langle H \rangle = -(\partial \ln \Delta / \partial \beta)_{N, P_{\text{ext}}}$, and $\langle V \rangle = (\partial G / \partial P_{\text{ext}})_{N, T}$. The enthalpy contains the imposed pressure P_{ext} , not a frame-dependent instantaneous pressure. Artificial thermostat and barostat energies are not part of E or H .

The measure matters when constructing an NPT algorithm. For fractional coordinates $\mathbf{s}_i = \mathbf{h}^{-1} \mathbf{R}_i$, the Cartesian position measure becomes $d^{3N} R = V^N d^{3N} s$. Thus the density in fractional positions and Cartesian momenta contains a factor V^N in addition to the Boltzmann weight. Allowing cell shape to fluctuate requires a specified cell measure and the appropriate extended-system corrections; simply propagating the cell toward a target pressure does not guarantee NPT sampling [11]. The flexible-cell calculation in this report uses a hydrostatic external pressure while permitting anisotropic strain.

NVE, NVT, and NPT constrain different variables and have different fluctuations even when their macroscopic averages approach one another in the thermodynamic limit. All three keep the chemical composition fixed. Equilibration at a specified gas chemical potential would instead require particle exchange, for example through a grand-canonical sampling method. Fixed-loading MD therefore does not determine an adsorption isotherm by itself.

2.1.4 Equipartition, temperature, and initial velocities

Canonical integration by parts gives the equipartition identity $\langle x_a \partial \mathcal{H} / \partial x_a \rangle = k_B T$ for an independent phase-space coordinate x_a with vanishing boundary terms. Each independent quadratic term contributes $k_B T/2$. For f active quadratic momentum degrees of freedom,

$$\langle K \rangle = \frac{f}{2} k_B T, \quad T_{\text{kin}} = \frac{2K}{f k_B}. \quad (14)$$

For unconstrained atoms $f = 3N$; subtract independent holonomic constraints and any deliberately removed momentum degrees of freedom. Fixing the total linear momentum removes three degrees of freedom. Overall rotations are not normally removed from a periodic bulk system. Thermostat and barostat auxiliary variables do not enter the physical kinetic temperature.

The Maxwell–Boltzmann distribution provides initial velocity components:

$$v_{i\alpha} \sim \mathcal{N}\left(0, \frac{k_B T}{m_i}\right), \quad \alpha \in \{x, y, z\}. \quad (15)$$

If zero total momentum is required, subtract $\mathbf{v}_{\text{cm}} = (\sum_i m_i \mathbf{v}_i) / (\sum_i m_i)$ and use the matching f . Constrained velocities must also satisfy the velocity constraints. One-time rescaling may set an initial kinetic energy, but repeated exact rescaling to a fixed K does not generate canonical kinetic-energy fluctuations. In the canonical ensemble, $\text{var}(K) = f(k_B T)^2/2$ and hence $\text{var}(T_{\text{kin}}) = 2T^2/f$. The target temperature fixes a distribution, not the temperature of every saved frame.

2.1.5 Periodic boundaries, interactions, and pressure

Periodic boundaries approximate bulk material by replicating the cell. Positions differing by $\mathbf{h}\mathbf{n}$, with $\mathbf{n} \in \mathbb{Z}^3$, represent periodic images. For short-range interactions, the minimum-image convention uses the nearest image of a pair. It is valid for a single-image cutoff only when r_c is smaller than half the shortest nonzero lattice-vector length; otherwise all images inside the cutoff must be included explicitly. For skew cells the nearest image requires the cell geometry, not independent rounding of Cartesian components. This condition must remain valid as an NPT cell deforms. Periodic boundaries remove surfaces but leave finite-size effects, so the cell size must be converged for the observables of interest.

A neighbor list stores candidate interactions out to $r_{\text{list}} = r_c + r_{\text{skin}}$. It must be rebuilt before particle motion or cell deformation can bring an omitted pair inside r_c ; at fixed cell, a common criterion limits each displacement since rebuilding to half the skin. Smooth cutoff treatment avoids force jumps and energy drift. Conventional long-range electrostatics requires a method such as Ewald summation or particle-mesh Ewald; a short cutoff alone is generally insufficient. Any separate long-range term must be consistent with the chosen potential so that interactions are not double counted.

Pressure contains both a kinetic contribution and an interaction virial. For an unconstrained system with peculiar velocities (velocities with any bulk streaming motion removed), an instantaneous pressure tensor is

$$\mathbf{P}_{\text{inst}} = \frac{1}{V} \left[\sum_i \frac{\mathbf{p}_i \otimes \mathbf{p}_i}{m_i} + \mathbf{W} \right], \quad P_{\text{inst}} = \frac{1}{3} \text{tr } \mathbf{P}_{\text{inst}}. \quad (16)$$

The outer product is denoted by $\otimes$. For pair forces, $\mathbf{W} = \sum_{i < j} \mathbf{r}_{ij} \otimes \mathbf{F}_{ij}$, where $\mathbf{r}_{ij} = \mathbf{R}_i - \mathbf{R}_j$ uses the appropriate periodic image and $\mathbf{F}_{ij}$ is the force on i due to j . For a many-body potential, the virial is obtained consistently from the response of the energy to a homogeneous strain at fixed fractional coordinates,

$$W_{\alpha\beta} = - \left. \frac{\partial U}{\partial \epsilon_{\alpha\beta}} \right|_{\epsilon=0, \{\mathbf{s}_i\}}, \quad \mathbf{h} \mapsto (\mathbf{I} + \epsilon) \mathbf{h}. \quad (17)$$

For the rotationally invariant potentials considered here the virial is symmetric. Constraint forces and long-range terms, when present, must also be included. Stress conventions vary: tensile-positive Cauchy stress has opposite sign to the compressive-positive pressure tensor. The potential virial returned by a model must also be distinguished from the full pressure, which includes kinetic motion [3].

2.1.6 Time integration and numerical accuracy

The continuous trajectory is approximated at times $t_n = n\Delta t$. For a separable, fixed-cell Hamiltonian, velocity Verlet consists of a half momentum update, a position update, force reevaluation, and a second half momentum update:

$$\mathbf{p}_i^{n+1/2} = \mathbf{p}_i^n + \frac{\Delta t}{2} \mathbf{F}_i(\mathbf{R}^n), \quad (18)$$

$$\mathbf{R}_i^{n+1} = \mathbf{R}_i^n + \Delta t \frac{\mathbf{p}_i^{n+1/2}}{m_i}, \quad (19)$$

$$\mathbf{p}_i^{n+1} = \mathbf{p}_i^{n+1/2} + \frac{\Delta t}{2} \mathbf{F}_i(\mathbf{R}^{n+1}). \quad (20)$$

After initialization, the old forces are reused and one new force evaluation is needed per step. The scheme is time reversible, symplectic, and globally second order in Δt . For smooth forces and sufficiently small steps it approximately conserves a nearby shadow Hamiltonian, giving bounded physical energy oscillations rather than exact conservation. Even the harmonic stability bound $\omega_{\max}\Delta t < 2$ is much weaker than the condition needed for accurate sampling. The timestep should resolve the fastest motion and be reduced until the target observables converge.

Bond constraints, if adopted, are enforced with algorithms such as SHAKE for positions and RATTLE for momenta. They can remove the fastest bond vibrations and permit larger steps, but they change the model's degrees of freedom and heat capacity. A constrained trajectory cannot be combined with an unconstrained harmonic correction without consistent mode bookkeeping. The fully flexible model in this report retains these vibrations.

A short NVE trajectory tests timestep, force continuity, precision, neighbor lists, and constraint tolerances through energy conservation. Compare energy drift and fluctuation amplitude at successively smaller timesteps; chaotic trajectories need not agree point by point. Thermostatted dynamics can conceal integration errors by removing the resulting heat. NVT and NPT integrators include additional thermostat and barostat substeps and must also be checked for finite-timestep bias in equilibrium averages and fluctuations.

2.1.7 Thermostats and barostats

A thermostat should generate the canonical distribution, including its fluctuations. For example, Langevin dynamics adds friction and a zero-mean Gaussian ran-

dom force:

$$\dot{\mathbf{p}}_i = \mathbf{F}_i - \gamma \mathbf{p}_i + \boldsymbol{\eta}_i(t), \quad (21)$$

$$\langle \eta_{i\alpha}(t) \eta_{j\beta}(t') \rangle = 2m_i \gamma k_B T \delta_{ij} \delta_{\alpha\beta} \delta(t - t'). \quad (22)$$

Here γ is an inverse time, the indexed deltas are Kronecker deltas, and the time delta is a Dirac delta. The noise amplitude and friction obey fluctuation-dissipation balance, making the canonical distribution stationary in continuous time. Weak friction exchanges heat slowly; strong friction can overdamp motion. A suitable stochastic integrator is required at finite Δt .

A deterministic alternative is the Nosé-Hoover thermostat. Its simplest physical-variable form introduces a friction variable ξ :

$$\dot{\mathbf{p}}_i = \mathbf{F}_i - \xi \mathbf{p}_i, \quad \dot{\xi} = \frac{2K - f k_B T}{Q}, \quad (23)$$

where the thermostat inertia Q sets a coupling timescale, typically through $Q \sim f k_B T \tau_T^2$. A single variable can sample poorly for nearly harmonic systems. Nosé-Hoover chains couple additional thermostat variables and improve sampling, although ergodicity still needs to be assessed [12]. Simple weak-coupling or exact velocity-rescaling schemes may stabilize a starting structure but generally suppress the canonical fluctuations required for fluctuation-based heat capacities.

A barostat additionally evolves volume or cell strain in response to the internal pressure relative to P_{ext} . Isotropic coupling changes volume at fixed shape; flexible-cell coupling also permits shear and unequal changes of the cell axes. Martyna-Tobias-Klein-type algorithms include the extended variables and measure corrections needed for isobaric sampling [11]. The pressure damping time controls how quickly the cell responds: overly rapid coupling can excite cell oscillations or destabilize a soft framework, whereas overly slow coupling reduces sampling efficiency. Instantaneous pressure is expected to fluctuate; the equilibrium mean should agree with the target within uncertainty.

Thermostat and barostat settings must be converged along with the timestep. Correct equilibrium averages do not imply correct time-dependent dynamics: stochastic forces and strong coupling can alter diffusion and correlation functions. Transport properties require a justified dynamical protocol, commonly an equilibrated fixed-cell NVE trajectory or demonstrably weak coupling. Heat capacities instead require correct equilibrium sampling of the chosen ensemble.

2.1.8 Fluctuations, response functions, and heat capacity

Partition-function derivatives explain why equilibrium fluctuations measure thermodynamic responses. For a canonical observable A with no explicit temperature dependence, differentiating both its Boltzmann weight and the normalization gives

$$\left(\frac{\partial \langle A \rangle}{\partial T} \right)_{N,V} = \frac{\text{cov}(A, E)}{k_B T^2}, \quad (24)$$

where $\text{cov}(A, B) = \langle AB \rangle - \langle A \rangle \langle B \rangle$. In NPT, the corresponding identity replaces E by $H = E + P_{\text{ext}}V$ and holds at fixed external pressure. Consequently,

$$C_V = \left(\frac{\partial \langle E \rangle}{\partial T} \right)_{N,V} = \frac{\text{var}_{NVT}(E)}{k_B T^2}, \quad (25)$$

$$C_P = \left(\frac{\partial \langle H \rangle}{\partial T} \right)_{N,P_{\text{ext}}} = \frac{\text{var}_{NPT}(H)}{k_B T^2}. \quad (26)$$

These are variances of the *physical* energy and enthalpy, excluding extended-system energies. In particular, the canonical energy-fluctuation formula cannot be applied to an NVE trajectory whose total energy is fixed. For separable Cartesian kinetic energy with a fixed number f of active quadratic momentum degrees of freedom, kinetic and configurational variables are independent, so equivalent estimators are

$$C_V = \frac{f}{2} k_B + \frac{\text{var}_{NVT}(U)}{k_B T^2}, \quad (27)$$

$$C_P = \frac{f}{2} k_B + \frac{\text{var}_{NPT}(U + P_{\text{ext}}V)}{k_B T^2}. \quad (28)$$

Using potential-energy fluctuations alone without the kinetic term would therefore omit part of the heat capacity.

Volume fluctuations and enthalpy–volume correlations similarly give

$$\kappa_T = -\frac{1}{\langle V \rangle} \left(\frac{\partial \langle V \rangle}{\partial P_{\text{ext}}} \right)_{N,T} = \frac{\text{var}_{NPT}(V)}{k_B T \langle V \rangle}, \quad (29)$$

$$\alpha_V = \frac{1}{\langle V \rangle} \left(\frac{\partial \langle V \rangle}{\partial T} \right)_{N,P_{\text{ext}}} = \frac{\text{cov}_{NPT}(V, H)}{k_B T^2 \langle V \rangle}. \quad (30)$$

For a stable single phase in hydrostatic equilibrium, the thermodynamic identity is

$$C_P - C_V = \frac{TV\alpha_V^2}{\kappa_T}, \quad (31)$$

where V is the equilibrium volume and other mechanical conditions must be consistent between the derivatives. The difference need not be negligible in a flexible porous solid. These response identities assume a temperature-independent Hamiltonian and fixed composition; explicitly state-dependent effective potentials require additional derivative terms.

An alternative to fluctuation estimators is to simulate several temperatures and differentiate their equilibrium mean energies or enthalpies. This is the route used here for the classical NPT term. Temperature spacing controls finite-difference bias, while trajectory length controls the uncertainty of each mean; making the spacing smaller can amplify noise in the derivative. Fluctuation estimates provide a useful cross-check when their longer sampling requirements are met. All quantities above are extensive heat capacities of the simulated cell. The corresponding mass-specific value is $c_P = C_P/M_{\text{cell}}$, with M_{cell} the total loaded-cell mass.

2.1.9 Equilibration, correlation, and statistical uncertainty

Initial coordinates and velocities are not generally equilibrium samples. Equilibration is the transient during

which the energy, density, structure, and guest distribution approach their stationary distributions. Its frames must be excluded from equilibrium estimates. Minimizing a structure only finds a nearby low-energy configuration; it neither samples finite-temperature entropy nor replaces equilibration. Running averages, block averages, structural observables, and comparisons between independently initialized replicas should guide the discarded interval.

For M production samples A_n saved at interval δt_s , define

$$\bar{A} = \frac{1}{M} \sum_{n=1}^M A_n, \quad (32)$$

$$C_A(k) = \langle (A_n - \langle A \rangle)(A_{n+k} - \langle A \rangle) \rangle, \quad (33)$$

$$\rho_A(k) = C_A(k)/C_A(0). \quad (34)$$

The normalized autocorrelation ρ_A describes memory between saved frames. For a stationary trajectory, the exact variance of its sample mean is

$$\text{var}(\bar{A}) = \frac{C_A(0)}{M} \left[1 + 2 \sum_{k=1}^{M-1} \left(1 - \frac{k}{M} \right) \rho_A(k) \right]. \quad (35)$$

For a run much longer than its correlation time, define the statistical inefficiency g_A and integrated autocorrelation time $\tau_{\text{int},A}$:

$$g_A = 1 + 2 \sum_{k=1}^{\infty} \rho_A(k), \quad \tau_{\text{int},A} = \frac{\delta t_s}{2} g_A, \quad (36)$$

$$M_{\text{eff},A} \simeq \frac{M}{g_A}, \quad \text{SE}(\bar{A}) \simeq \sqrt{\frac{C_A(0)}{M_{\text{eff},A}}}. \quad (37)$$

The infinite sum must be truncated or estimated with a statistically justified procedure; noisy long-lag correlations cannot simply be summed indefinitely. The effective sample count depends on the observable. Saving frames more frequently does not create additional independent information over a fixed physical duration.

Block averaging is a practical alternative: divide the stationary trajectory into blocks longer than the relevant correlation time and estimate the error from the block means. The estimated error should stabilize as block length increases while enough blocks remain for a meaningful estimate. Blocking methods for correlated data are discussed by Flyvbjerg and Petersen [4]. Independent replicas test reproducibility and can expose slow modes missed within one trajectory. Neither blocking nor a small error bar can diagnose an important basin that every replica fails to visit. Uncertainty in a variance-based heat capacity also depends on correlations of the squared fluctuations and is not given by the standard error of the mean enthalpy alone.

For derivatives and combined estimators, resample independent replicas or sufficiently long blocks and repeat the entire analysis, preserving any cross-temperature correlations. Resampling individual correlated frames as if independent underestimates uncertainty. Sampling error decreases with more independent data; finite-size error, timestep bias, potential-model error, and missing nuclear quantum effects require separate convergence or validation checks.

2.1.10 Complete simulation workflow

The preceding theory leads to the following sequence for an equilibrium MD calculation:

1. **Define the physical question and ensemble.** Specify the material, composition, loading, target observables, temperature, and either volume or external pressure. Decide whether cell shape may change and whether any constraints are part of the physical model. Fixed-loading NPT is appropriate for the present enthalpy curve.
2. **Choose and validate the interaction model.** Assign species, masses, units, and any required charges or topology. Check energies and forces for representative configurations and validate stresses if the cell will evolve. Establish cutoff, long-range treatment, and the model’s applicable temperature and structural range.
3. **Build the initial configuration.** Choose a cell or supercell, apply periodic boundaries, insert guests at the desired loading, reject severe overlaps, and inspect geometry and composition. Remove large residual forces by minimization if needed. Generate independent initial placements for replicas; random insertion is not an equilibrium adsorption distribution.
4. **Initialize motion and numerical parameters.** Draw velocities, impose the intended momentum and constraint conditions, and select the integrator, timestep, neighbor-list settings, coupling times, random seeds, trajectory stride, and restart frequency. Evaluate the initial energy and forces before taking the first step. Validate energy conservation in short NVE trials at several timesteps.
5. **Equilibrate at the target conditions.** Couple to the required reservoirs, using gradual heating or staged volume relaxation when necessary. Monitor enthalpy, temperature, pressure, volume, framework integrity, and guest motion. Continue until the relevant observables are stationary and replicas give compatible distributions.
6. **Collect production trajectories.** Keep the physical model and target ensemble fixed. At each step update positions and, in NPT, the cell; update periodic images and neighbor lists; evaluate energies, forces, and virials; and finish the momentum and reservoir updates in the order required by the integrator. Save physical observables and configurations at the chosen stride, together with restart states.
7. **Analyze and quantify uncertainty.** Discard equilibration, compute ensemble averages and correlations, estimate effective sample counts or block errors, and combine independent replicas. Unwrap periodic trajectories when analyzing displacements, accounting for cell motion in NPT. For this report, differentiate the mean physical enthalpy

with respect to temperature and propagate its sampling error.

8. **Converge and document the result.** Check timestep, system size, interaction settings, coupling times, production length, replica count, and temperature spacing against the desired accuracy. Record the model identity, input structures, software versions, seeds, ensemble parameters, discarded interval, and analysis procedure so that the reported heat capacity can be reproduced.

The machine-learned potential introduced next supplies U , its forces, and its cell derivatives in this general framework. The later methodology section specifies the actual MOF-5 settings; the harmonic correction addresses the quantum populations that classical MD cannot sample.

2.2 Machine-learned interatomic potentials

2.2.1 Learning the potential-energy surface

Within the Born–Oppenheimer approximation, the electronic ground-state energy defines a potential-energy surface (PES) for the nuclei. An interatomic potential replaces an electronic-structure calculation by an approximation to this map,

$$(\mathbf{R}, \mathbf{h}, \mathbf{Z}) \mapsto U(\mathbf{R}, \mathbf{h}, \mathbf{Z}), \quad (38)$$

where $\mathbf{R} = (\mathbf{R}_1, \dots, \mathbf{R}_N)$ are the nuclear coordinates, $\mathbf{h}$ is the periodic cell, and $\mathbf{Z}$ lists the chemical species. A machine-learned interatomic potential (MLIP) represents this map by a differentiable function with parameters θ , fitted to energies and forces from electronic-structure calculations. Modern local MLIPs usually write the extensive energy as

$$U_\theta(\mathbf{R}, \mathbf{h}, \mathbf{Z}) = U_{\text{base}}(\mathbf{Z}) + \sum_{i=1}^N \varepsilon_i(\mathcal{N}_i; \theta), \quad (39)$$

where U_{base} is an optional composition-dependent baseline and $\mathcal{N}_i$ is the learned environment of atom i . The decomposition makes the cost approximately linear in N at fixed density and makes the energy size extensive. The individual atomic energies are a model construction rather than uniquely defined physical observables.

The force field is obtained by differentiating the scalar energy,

$$\mathbf{F}_i = -\frac{\partial U_\theta}{\partial \mathbf{R}_i}, \quad (40)$$

so the predicted forces are conservative, provided the same differentiable energy model is used throughout. Cell derivatives similarly give the virial or stress when the model and its interface support them. Automatic differentiation can be applied a second time to obtain the Cartesian Hessian used later in this project.

2.2.2 From an atomic structure to a graph

A graph neural network (GNN) acts on an attributed graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, following the notation of Bronstein

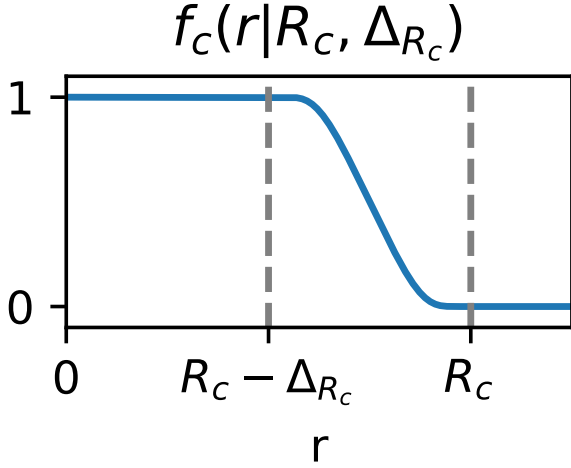


Figure 1: A smooth radial cutoff suppresses a neighbor contribution as its distance approaches the outer cutoff r_c . Smooth decay is needed for continuous energies, forces, and Hessians. Figure from the PET paper [9].

et al. [5]. The node set $\mathcal{V}$ contains the $N = |\mathcal{V}|$ atoms. An input node signal is a map $\mathbf{x} : \mathcal{V} \rightarrow \mathbb{R}^s$; its value at node u is $\mathbf{x}_u$, and stacking all node features as rows gives $\mathbf{X} = (\mathbf{x}_1, \dots, \mathbf{x}_N)^T \in \mathbb{R}^{N \times s}$. The edge set encodes local interactions. Because distinct periodic images of one atom may enter the same environment, the atomistic graph is more precisely a directed periodic multigraph. An edge $(v, u, \mathbf{n}) \in \mathcal{E}$, directed from the image $(v, \mathbf{n})$ to the central atom u , is included when

$$\mathbf{r}_{uv\mathbf{n}} = \mathbf{R}_v + \mathbf{h}\mathbf{n} - \mathbf{R}_u, \quad (41)$$

$$r_{uv\mathbf{n}} = \|\mathbf{r}_{uv\mathbf{n}}\| < r_c, \quad (42)$$

where $\mathbf{n} \in \mathbb{Z}^3$ identifies a periodic image and r_c is a cutoff. The incoming one-hop neighborhood is therefore the indexed set

$$\mathcal{N}_u = \{(v, \mathbf{n}) : (v, u, \mathbf{n}) \in \mathcal{E}\}. \quad (43)$$

This is the periodic analogue of the neighborhood $\mathcal{N}_u$ in Ref. [5]; retaining $\mathbf{n}$ distinguishes repeated images of the same node. Relative displacements remove dependence on the absolute origin. A sufficiently smooth cutoff function $f_c(r)$, whose value and relevant derivatives vanish at r_c , suppresses a neighbor’s contribution as it leaves the environment. Smooth forces and Hessians require consistent cutoff treatment throughout the network, including attention normalization and energy readout, rather than continuity of the cutoff value alone. The representative single cutoff and the comparison of cutoff functions are shown in Figures 1 and 2, respectively; both are from the PET paper [9].

Neural networks operate on fixed-width numerical representations. Chemical identity is therefore mapped from a discrete species label to a trainable embedding vector,

$$\mathbf{x}_u^{(0)} = \mathbf{E}_Z[Z_u] \in \mathbb{R}^d, \quad (44)$$

where each row of the embedding matrix $\mathbf{E}_Z$ is learned during training. In the attributed-graph notation, geometry is carried by edge features. For example, a

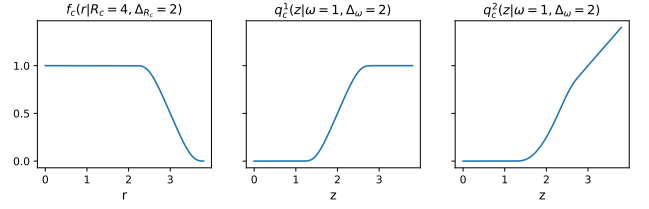


Figure 2: Comparison of radial cutoff functions used to attenuate neighbor contributions near r_c . The value and the derivatives relevant to the model must be treated consistently in every energy contribution. Figure from the PET paper [9].

radial edge signal can be written

$$\mathbf{x}_{uv\mathbf{n}}^{\text{edge}} = f_c(r_{uv\mathbf{n}})[\varphi_1(r_{uv\mathbf{n}}), \dots, \varphi_K(r_{uv\mathbf{n}})], \quad (45)$$

and, in models that retain directional information, by the displacement $\mathbf{r}_{uv\mathbf{n}}$, its unit vector, spherical harmonics, or learned functions of these quantities. Here “embedding vector” means a vector in latent feature space; it need not transform as a physical three-dimensional vector.

Suppressing periodic-image labels for a moment, graph connectivity may also be represented by an adjacency matrix $\mathbf{A}$. A relabeling of the nodes by a permutation matrix $\mathbf{P}$ sends $(\mathbf{X}, \mathbf{A})$ to $(\mathbf{P}\mathbf{X}, \mathbf{P}\mathbf{A}\mathbf{P}^T)$. The geometric deep-learning prescription is that a node-wise GNN F be permutation equivariant, whereas a graph-wise scalar readout f be permutation invariant:

$$F(\mathbf{P}\mathbf{X}, \mathbf{P}\mathbf{A}\mathbf{P}^T) = \mathbf{P}F(\mathbf{X}, \mathbf{A}), \quad (46)$$

$$f(\mathbf{P}\mathbf{X}, \mathbf{P}\mathbf{A}\mathbf{P}^T) = f(\mathbf{X}, \mathbf{A}). \quad (47)$$

For an interatomic potential, the first relation applies to node representations and the second to the total energy. Periodic-image and edge features must be relabeled consistently with their incident nodes.

2.2.3 Message passing and many-body representations

In the notation of Ref. [5], a message-passing GNN computes a latent node feature $\mathbf{h}_u$ by applying a learnable message function ψ on every incoming edge, combining the messages with a permutation-invariant aggregation $\bigoplus$, and applying a learnable update function ϕ :

$$\mathbf{h}_u = \phi \left(\mathbf{x}_u, \bigoplus_{(v, \mathbf{n}) \in \mathcal{N}_u} \psi(\mathbf{x}_u, \mathbf{x}_v, \mathbf{x}_{uv\mathbf{n}}^{\text{edge}}) \right). \quad (48)$$

Equation (48) extends the book’s node-feature equation by conditioning ψ on the geometric edge feature, as required for an attributed atomistic graph. A sum is the usual choice for $\bigoplus$, though means, maxima, and other symmetric multiset functions are possible. For a deep GNN, take $\mathbf{h}_u^{(0)} = \mathbf{x}_u^{(0)}$, apply Eq. (48) recursively with layer-specific ψ_ℓ and ϕ_ℓ , and denote each layer’s output by $\mathbf{h}_u^{(\ell+1)}$. Although the graph initially contains only pairwise edges, repeated updates create

many-body features: after L layers, an atom can depend on paths extending up to L graph hops. A shared readout ρ maps the final node features to local energies, $\varepsilon_u = \rho(\mathbf{h}_u^{(L)})$, whose sum gives the invariant extensive energy in Eq. (39).

2.2.4 Point Edge Transformer architecture

The Point Edge Transformer (PET) introduced by Pozdnyakov and Ceriotti [9] is the architecture underlying the later PET-MAD potential [8]. PET preserves a distinct latent message on each directed edge. For a central atom i , its transformer processes the incoming edge messages together and produces a separate outgoing message for each neighbor. This avoids compressing the entire neighborhood into a single atom representation before forming outgoing messages. Attention still performs weighted sums internally; the distinction is that the edge identities survive the update.

In the original formulation (Ref. [9], Appendix A), neighbor species, relative Cartesian position, and an incoming message $\mathbf{m}_{j \rightarrow i}^{(\ell-1)} \in \mathbb{R}^{d_{\text{PET}}}$ are encoded as

$$\mathbf{p}_{ij}^{(\ell)} = \text{SiLU}(\mathbf{W}_r^{(\ell)} \mathbf{r}_{ij} + \mathbf{b}_r^{(\ell)}), \quad (49)$$

$$\mathbf{t}_{ij}^{(\ell)} = \text{MLP}_\ell(\mathbf{m}_{j \rightarrow i}^{(\ell-1)} \parallel \mathbf{E}_n^{(\ell)}[Z_j] \parallel \mathbf{p}_{ij}^{(\ell)}), \quad (50)$$

where $\parallel$ denotes concatenation. Each ingredient has width d_{PET} , and the MLP compresses their concatenation from $3d_{\text{PET}}$ to d_{PET} . In the first block the incoming message is absent, giving $2d_{\text{PET}}$ input features. A separate central token, $\mathbf{t}_{i0}^{(\ell)} = \mathbf{E}_c^{(\ell)}[Z_i]$, uses a distinct species embedding. These are latent feature vectors with no prescribed rotation law. Later PET implementations can modify the encoding and feature dimensions; the project-specific settings are given in Section 4.2.

A transformer mixes the central and neighbor tokens within one environment. For one attention head, with a, b indexing these tokens,

$$\mathbf{q}_a = \mathbf{W}_Q \mathbf{t}_a, \quad \mathbf{k}_b = \mathbf{W}_K \mathbf{t}_b, \quad (51)$$

$$\mathbf{v}_b = \mathbf{W}_V \mathbf{t}_b, \quad (52)$$

$$\alpha_{ab} = \frac{\exp(\mathbf{q}_a^T \mathbf{k}_b / \sqrt{d_k}) c_b}{\sum_c \exp(\mathbf{q}_a^T \mathbf{k}_c / \sqrt{d_k}) c_c}, \quad (53)$$

$$\mathbf{y}_a = \sum_b \alpha_{ab} \mathbf{v}_b. \quad (54)$$

where $c_0 = 1$ for the central token and $c_j = f_c(r_{ij})$ for neighbor j . The cutoff enters before normalization, so a departing neighbor vanishes smoothly from both the weighted sum and its denominator. Multi-head attention repeats this operation in learned subspaces, followed by feed-forward layers and residual connections. No arbitrary neighbor order enters the calculation, so permuting neighbors permutes their output tokens consistently.

Attention makes each neighbor token depend on the other neighbors, allowing many-body interactions within a single message-passing block. Adding attention layers increases local expressive power without ex-

tending that block’s neighborhood. Adding message-passing blocks communicates information between environments and enlarges the receptive field. These are separate architecture choices: the water benchmark in Ref. [9] shows that deeper local transformers can outperform a larger number of shallow message-passing blocks at fixed total attention-layer count. Local attention is quadratic in the number of tokens per environment, while the total cost remains approximately linear in atom count for a fixed architecture and bounded neighborhood size.

The output neighbor tokens supply outgoing messages, updated with residual connections to preceding messages. Each block also contributes to the energy. For the summed-edge variant in the original paper,

$$U_\theta = \sum_i b_{Z_i} + \sum_{\ell=1}^L \sum_i \left[H_c^{(\ell)}(\mathbf{x}_{i0}^{(\ell)}) + \sum_{j \in \mathcal{N}_i} f_c(r_{ij}) H_n^{(\ell)}(\mathbf{x}_{ij}^{(\ell)}) \right], \quad (55)$$

where $\mathbf{x}_{i0}$ and $\mathbf{x}_{ij}$ are central and neighbor output tokens, H_c and H_n are learned readout heads, and b_Z are fitted species baselines. Appendix A of Ref. [9] also describes a cutoff-weighted average of edge predictions. The edge terms are many-body learned contributions, not physical pair energies. Contributions from all blocks enter the total, and cutoff weighting is required in the readout as well as in attention.

2.2.5 Physical symmetries

For an isolated non-magnetic system without external fields, a valid scalar energy should be unchanged by translation, rotation or reflection of the whole structure and by permutation of identical atoms:

$$U(\mathbf{R}) = U(\mathbf{R} + \mathbf{a}) = U(\mathbf{Q}\mathbf{R}) = U(\mathbf{P}\mathbf{R}), \quad \mathbf{Q} \in O(3). \quad (56)$$

An invariant scalar is unchanged by a symmetry operation, whereas an equivariant vector changes in the corresponding way; for example, $\mathbf{F}_i(\mathbf{Q}\mathbf{R}) = \mathbf{Q}\mathbf{F}_i(\mathbf{R})$. Translation invariance follows from using relative positions, and the shared local functions and symmetric aggregation enforce permutation invariance. Many MLIPs impose rotational symmetry exactly through invariant geometric descriptors or $E(3)$ -equivariant features. PET instead processes Cartesian displacement vectors with an unconstrained transformer and learns rotational behavior from random rotational augmentation [9, 8]. Thus the PET-MAD backbone is not intrinsically exactly rotation invariant. The PET-MAD publication reports rotational discrepancies usually smaller than its prediction errors [8]; this is not a validation of the particular MOF-5 structures or checkpoint used here. For periodic structures, rotation tests must rotate both positions and cell vectors.

2.2.6 Exact rotational symmetrization with ECSE

The original PET paper also introduces the Equivariant Coordinate System Ensemble (ECSE), a separate

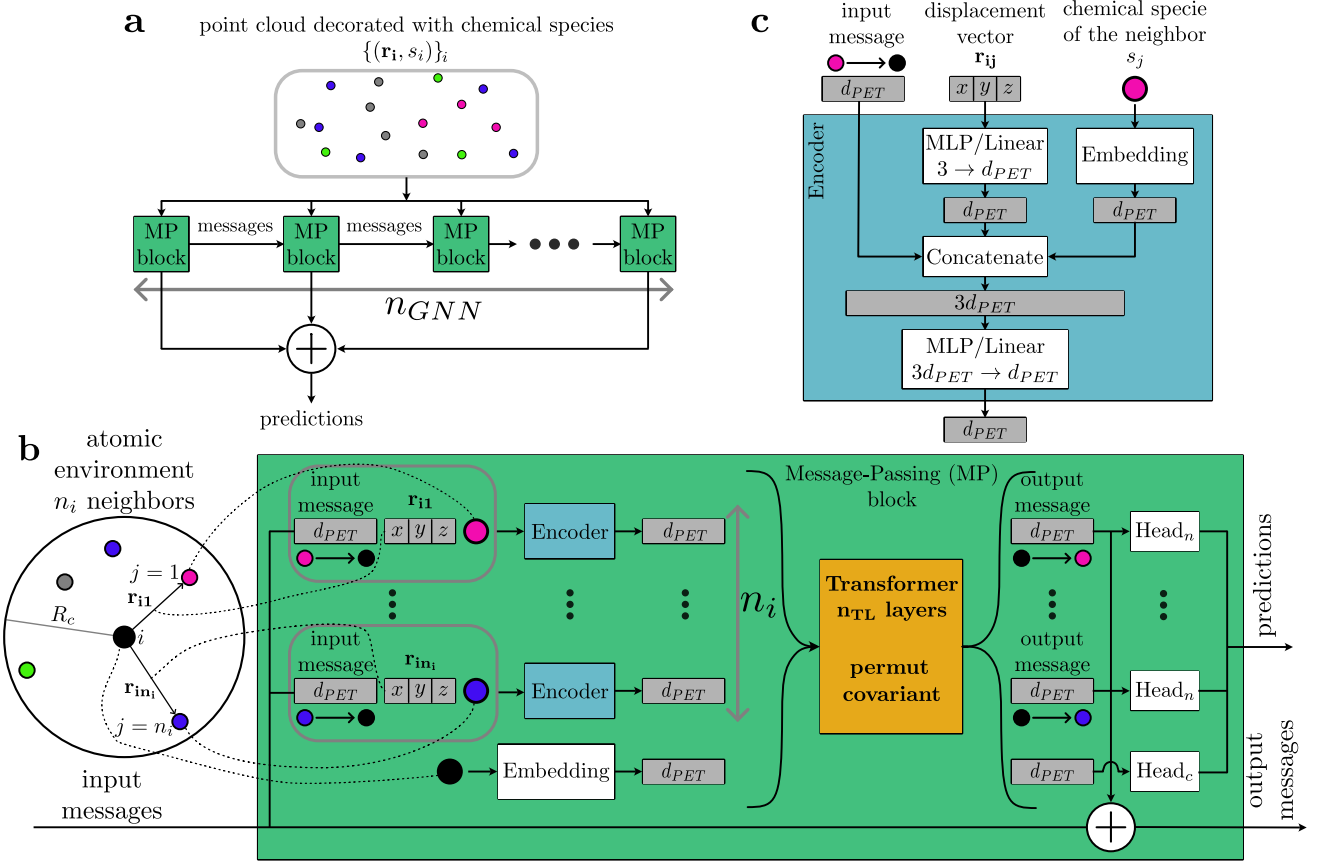


Figure 3: Point Edge Transformer (PET) message-passing architecture. Edge messages associated with a central atom are jointly processed by attention while retaining their individual edge identities for the outgoing update. Figure from the PET paper [9].

construction that imposes exact rotational symmetry on a smooth, translation- and permutation-invariant backbone [9]. Each ordered noncollinear neighbor pair (j, k) defines a local frame $\mathbf{Q}_{ijk}$ that rotates with the atomic environment. For a scalar local prediction, the symmetrized model is

$$\varepsilon_i^{\text{ECSE}} = \frac{\sum_{j \neq k} w_{ijk} \varepsilon_i^0(\mathbf{Q}_{ijk}^T \mathcal{N}_i)}{\sum_{j \neq k} w_{ijk}}, \quad (57)$$

where the notation indicates expressing all relevant displacement vectors in that frame. Vector or tensor predictions are transformed back to the original frame before averaging. The frames in the paper use proper rotations, $SO(3)$; reflection invariance in Eq. (56) is an additional requirement, not a consequence of this construction alone.

Radial weights suppress frames whose defining neighbors approach the cutoff, and angular weights suppress nearly collinear pairs. An adaptive inner radius and smooth pruning reduce the number of frames. Fully collinear environments require a fallback; the paper demonstrates an auxiliary invariant model. Using a smooth ensemble avoids the discontinuities caused by selecting only the two nearest neighbors. This frame-selection radius is distinct from the adaptive neighbor cutoff in the later PET implementation used here.

Forces and Hessians must differentiate the full expression in Eq. (57), including the position dependence of frames and weights. They cannot be obtained simply

by averaging derivatives of the backbone. The paper shows that tight frame-selection parameters can amplify derivative errors even when the energy remains mathematically smooth. Its proof-of-principle ECSE implementation incurs roughly three orders of magnitude of inference overhead (Ref. [9], Section 6 and Appendix F). ECSE is not enabled in this project’s workflow; its exact symmetry properties must therefore be distinguished from those of the deployed PET backbone.

2.2.7 Training and domain of validity

Given reference configurations s , a typical energy–force training objective is

$$\mathcal{L}(\theta) = \sum_s \left[w_E \left(\frac{U_\theta^{(s)} - U_{\text{ref}}^{(s)}}{N_s} \right)^2 + \frac{w_F}{3N_s} \sum_{i\alpha} \left(F_{i\alpha, \theta}^{(s)} - F_{i\alpha, \text{ref}}^{(s)} \right)^2 \right], \quad (58)$$

possibly augmented by stress terms and regularization. Because the force labels are derivatives of the PES, each configuration supplies $3N_s$ local slope constraints in addition to its energy. Optimization adjusts the embeddings, message-passing transformations and readout weights to minimize this loss. The original PET experiments first subtract fitted species baselines and use a loss that rescales energy and force er-

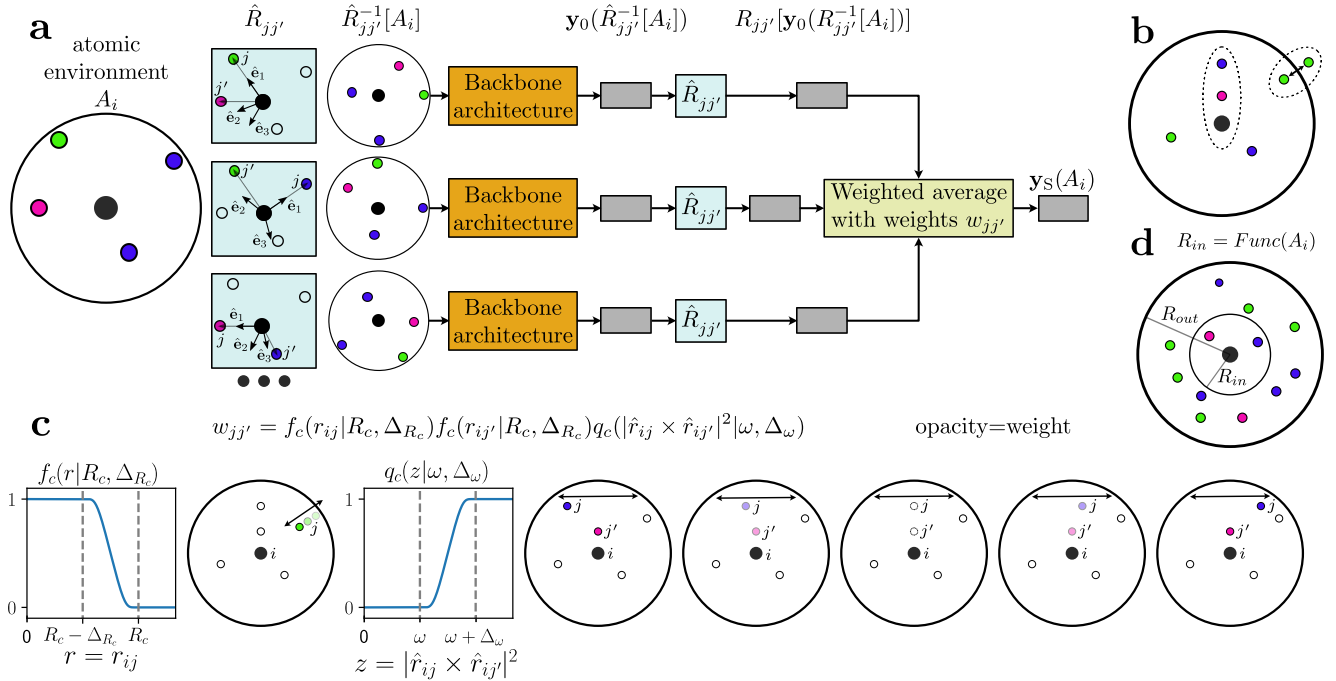


Figure 4: Equivariant Coordinate System Ensemble (ECSE). Local coordinate frames constructed from non-collinear neighbor pairs rotate with the atomic environment; predictions evaluated in these frames are transformed as needed and smoothly averaged. Figure from the PET paper [9].

rors by moving validation-set mean squared errors, together with Adam and random rotational augmentation (Ref. [9], Appendix B). Equation (58) is a generic objective, not that paper’s exact normalization. PET-MAD was trained on a chemically diverse DFT dataset rather than on MOF-5 alone, making it a universal pre-trained potential [8].

An MLIP interpolates the reference PES represented by its training data; it does not replace the approximations of the underlying electronic-structure method and need not extrapolate reliably to unfamiliar bonding, charge states, very short distances or thermodynamic phases. Energy and force test errors, trajectory stability and uncertainty indicators are therefore necessary but distinct checks. This project also uses second derivatives, which can be more sensitive than energies or forces and were not directly constrained by the usual loss in Eq. (58). Curvatures and vibrational spectra must therefore be validated explicitly. The original PET benchmarks include isolated distorted methane, water, molecular collisions, crystals, and high-entropy alloys, but do not validate methane adsorption in MOF-5. The paper also reports weaker high-temperature alloy extrapolation despite strong hold-out accuracy, illustrating the distinction between expressiveness and transferability [9].

2.3 Metal–organic frameworks and MOF-5

Metal–organic frameworks are crystalline networks assembled from metal ions or metal–oxo clusters joined by organic molecules called linkers. The connections repeat in three dimensions and leave accessible pores

between them. Changing the nodes and linkers can alter the pore size, surface chemistry, and framework flexibility. This makes MOFs useful for studying gas storage and separation: guest molecules can enter the pores and adsorb on their internal surfaces without becoming part of the framework.

MOF-5 provides a concrete example (Figure 5). Zinc–oxo clusters are connected by benzene-1,4-dicarboxylate linkers to form an open, periodic framework. Methane occupies its pores and interacts with the framework mainly through weak noncovalent forces. At a given temperature and loading, the methane molecules can move among adsorption environments, while the linkers and clusters also vibrate. The framework therefore supplies both the space that holds the gas and the flexible surroundings that influence its motion.

These motions explain why heat capacity is an informative property for this project. Heating excites framework vibrations and methane’s internal modes, but also changes how methane explores the pores and interacts with the host. The latter effects can be anharmonic and cannot be inferred from vibrations about a single structure [6]. We therefore use classical MD of *loaded* MOF-5 to sample the temperature-dependent host–guest motion, and normal modes of optimized loaded structures to correct the classical treatment of quantized vibrations. An empty MOF-5 Hessian provides a framework reference; the loaded-system Hessian supplies the correction used for the reported loaded-system heat capacity.

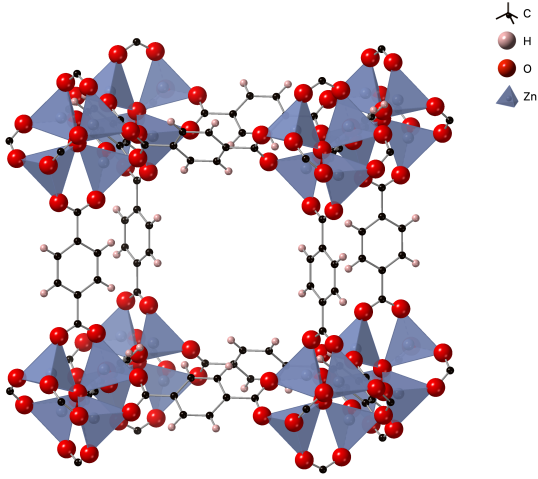


Figure 5: Ball-and-polyhedron representation of the MOF-5 framework. The zinc-oxo clusters are shown as blue polyhedra, connected by benzene-1,4-dicarboxylate linkers. The legend identifies carbon (black), hydrogen (pink), oxygen (red), and zinc (blue). The open spaces between framework components form the pores that host adsorbed molecules.

2.4 Thermodynamic definitions

2.4.1 Heat-capacity estimators used in this report

Section 2.1.8 derives the thermodynamic definitions and fluctuation formulas for C_V and C_P . The present workflow obtains the classical constant-pressure term by differentiating replica-averaged physical enthalpies $\langle H \rangle = \langle E + P_{\text{ext}}V \rangle$ on a temperature grid. The ensemble, loading, and external pressure are held fixed across that grid, and uncertainty is propagated through the differentiation. Equilibrated NPT enthalpy fluctuations provide an independent consistency check when sampling is sufficient.

The quantum correction is evaluated separately from harmonic normal-mode frequencies. It is formally a C_V correction, whereas the trajectory term is C_P . Equation (31) explains why adding these terms requires the pressure-volume contribution to the quantum correction to be small; the two heat capacities are not interchangeable by definition. Extensive cell heat capacities are combined before division by the total loaded-cell mass.

2.4.2 LLPR model uncertainty and CEA propagation

The uncertainty of a machine-learned potential is different from the sampling error of a finite MD trajectory. This project estimates the former for the classical NPT term and harmonic correction with a persistent last-layer prediction-rigidity (LLPR) ensemble [13], and the latter with autocorrelation-aware statistics and independent replicas. These quantities are kept separate

throughout the analysis.

For a fixed trained representation, write the central PET energy as $\bar{U}(\mathbf{x}) = \mathbf{w}_0^T \mathbf{f}(\mathbf{x})$, where $\mathbf{f}$ is the last-layer feature vector. If $\mathbf{F}$ contains the corresponding features of a reference-labeled covariance set, the meta-train LLPR standard deviation has the form

$$\sigma_{\text{LLPR}}^2(\mathbf{x}) = \alpha^2 \mathbf{f}(\mathbf{x})^T \times (\mathbf{F}^T \mathbf{F} + \lambda \mathbf{I})^{-1} \mathbf{f}(\mathbf{x}). \quad (59)$$

The regularizer λ stabilizes weakly constrained feature directions. The scale α is fitted against energy residuals on a separate, reference-labeled calibration set. The two sets must be disjoint and use the same electronic-structure definition as the base potential. Model predictions cannot serve as reference labels for calibrating their own errors.

Equation (59) is exported as `energy_uncertainty`, a standard deviation in energy units. That scalar is useful for calibration and out-of-domain diagnostics but cannot be propagated through an equilibrium average by itself. The propagation needs signed, correlated changes in energy across configurations. Metatraining therefore samples persistent shallow readout weights,

$$\mathbf{w}_i \sim \mathcal{N}[\mathbf{w}_0, \alpha^2 (\mathbf{F}^T \mathbf{F} + \lambda \mathbf{I})^{-1}], \quad (60)$$

$$U^{(i)}(\mathbf{x}) = \mathbf{w}_i^T \mathbf{f}(\mathbf{x}),$$

and exports their energies as `energy_ensemble`. In the large-member limit, their framewise standard deviation approaches the analytical LLPR standard deviation. This is a shallow ensemble: the learned representation is fixed, so it cannot reveal a model bias or representation error shared by all members and should not be described as a committee of independently trained neural networks.

The ensemble is propagated using the reweighting framework of Imbalzano *et al.* [15]. On a production frame t sampled by the central model at temperature T , define

$$\Delta U_t^{(i)} = U_t^{(i)} - \bar{U}_t, \quad (61)$$

$$H_t^{(i)} = K_t + U_t^{(i)} + P_{\text{ext}} V_t, \quad (62)$$

where K_t and cell volume V_t come from the central NPT trajectory. Normalized exponential reweighting gives

$$\langle H^{(i)} \rangle_i^{\text{direct}} = \frac{\sum_t e^{-\beta \Delta U_t^{(i)}} H_t^{(i)}}{\sum_t e^{-\beta \Delta U_t^{(i)}}}, \quad (63)$$

but can collapse when the energy difference is extensive. The primary large-system estimate is therefore the first-order cumulant expansion approximation (CEA),

$$\begin{aligned} \langle H^{(i)} \rangle_i^{\text{CEA}} &= \langle H^{(i)} \rangle_{\bar{U}} \\ &\quad - \beta \text{cov}_{\bar{U}}(H^{(i)}, \Delta U^{(i)}), \quad (64) \\ \beta &= (k_B T)^{-1}. \end{aligned}$$

Direct reweighting is retained as a diagnostic. Its member-wise effective sample count $N_{\text{eff}} = 1 / \sum_t \tilde{w}_t^2$

detects weight collapse. The additional diagnostic $\text{var}_t(\beta\Delta U_t^{(i)})$ should be much smaller than one for a controlled first-order CEA. A large value warns that omitted higher-order terms may be important; agreement with direct reweighting is meaningful only where N_{eff} remains adequate.

At each temperature, independent replicas are averaged member by member. The same exported ensemble and member index must be used at every temperature. The analysis then differentiates each complete enthalpy curve,

$$C_{P,\text{cl}}^{(i)}(T) = \frac{d\langle H^{(i)} \rangle_i^{\text{CEA}}}{dT},$$

$$\sigma_{\text{LLPR},C_P}(T) = \left\{ \frac{1}{M-1} \sum_{i=1}^M \left[C_{P,\text{cl}}^{(i)}(T) - \overline{C}_{P,\text{cl}}(T) \right]^2 \right\}^{1/2}. \quad (65)$$

The reported LLPR error is therefore the sample standard deviation of the final member-resolved C_P curves. It is not the instantaneous energy spread, and it is not divided by $\sqrt{M}$: ensemble members sample a predictive distribution rather than provide repeated measurements of one mean.

2.4.3 Gaussian NPT propagation of shallow-ensemble uncertainty

Direct propagation of shallow ensembles (DPOSE) carries persistent model members through a property calculation before taking their spread [14]. The saved comparison uses a project-specific Gaussian NPT approximation, rather than separate trajectories under each member. Define $Q_i = U_i + P_{\text{ext}}V$ and $\Delta U_i = U_i - U_0$, where subscript 0 identifies the central potential. If $(Q_i, \Delta U_i)$ is jointly Gaussian, exponential reweighting shifts its mean but leaves its covariance unchanged. Integrating out separable quadratic momenta then gives

$$C_{P,i}^G \simeq \frac{f}{2}k_B + \frac{\text{var}_0(Q_i)}{k_B T^2}. \quad (66)$$

The variance is measured on the central trajectory, not a member-equilibrated trajectory. The Gaussian assumption makes that substitution possible; it is not needed for the exact central-model fluctuation estimator in Section 2.1.8. Here $f = 3N - 3$, and extensive results are divided by the loaded-cell mass. Member indices must remain paired when adding the harmonic corrections so that classical-harmonic covariance is retained.

Two comparisons must be distinguished. The main Gaussian/DPOSE plots retain the central enthalpy-derivative curve and its sampling error, replacing only the propagated model spread. The separate variance-based hybrid calculation also replaces the central classical value and estimates its sampling error from the squared configurational-enthalpy fluctuations. Neither construction guarantees a smaller band or removes the need to validate sampling and the Gaussian approximation.

2.4.4 Why classical and quantum predictions differ

In the classical harmonic limit, every quadratic vibrational degree of freedom contributes k_B to C_V , producing the Dulong-Petit limit at all temperatures. Quantum mechanically, a mode of angular frequency ω becomes thermally populated only when $\hbar\omega$ is comparable to or smaller than $k_B T$. High-frequency framework and molecular modes are therefore partially frozen near room temperature. Classical MD can still describe anharmonic structural motion, but it cannot generate the correct quantum populations simply by extending the trajectory.

3 Reference evidence and the hybrid approximation

Kapil *et al.* studied empty MOF-5 and cells containing 50, 100, and 150 methane molecules [6]. Their results establish the physical motivation for combining a classical anharmonic term with a harmonic quantum correction, and test that approximation for their force field. This project uses the hybrid method as a computational strategy for PET, rather than as a recipe for reproducing their numerical results.

3.1 Differences from the reference calculation

The reference work used analytical QuickFF force fields derived from electronic-structure calculations, not PET-MAD [10]. Framework covalent terms were fitted to B3LYP cluster calculations, with MBIS charges obtained from a periodic PBE density. Nonbonded interactions combined Gaussian-distributed electrostatics with an MM3 Buckingham model and a 15 Å cutoff with tail corrections. Methane was inserted with RASPA, configurations with unrealistic contacts were rejected, and canonical Monte Carlo was used before dynamics. The present workflow instead uses PET for all energy, force, and Hessian evaluations, and independently generated loaded configurations. These differences are expected and are not defects to be removed by matching individual reference parameters.

The reference heat-capacity curves were evaluated from neighboring enthalpies,

$$C_P(T) \simeq \frac{\langle H \rangle_{T+\Delta T} - \langle H \rangle_{T-\Delta T}}{2\Delta T}, \quad (67)$$

$$\Delta T = 25 \text{ K},$$

rather than from a single short trajectory. The present classical term uses the same thermodynamic principle: every temperature point depends on converged, statistically independent simulations at adjacent temperatures.

For the harmonic comparison, optimized structures were used to construct Cartesian Hessians with Yaff and normal modes with TAMkin. Empty MOF-5 was tested with both QuickFF and UFF4MOF. The loaded 100-methane calculation used five different

Monte Carlo configurations, each optimized into a local minimum before its harmonic spectrum was evaluated.

3.2 Physical conclusions relevant to this project

The main findings of the reference study are:

- Empty MOF-5 is quantum mechanical but nearly harmonic over the studied range. Its quantum harmonic C_V is close to the reported C_P , with a room-temperature gravimetric value of approximately 0.7 J/g/K.
- Classical MD overestimates the empty-framework heat capacity because it populates all vibrational modes according to equipartition.
- Methane loading introduces strongly anharmonic host-guest motion. The reported heat-capacity curves develop a loading-dependent minimum around 200 K, caused mainly by changes in attractive host-guest interactions as confined methane becomes more mobile.
- A harmonic calculation cannot represent diffusion, cage-to-cage motion, or the temperature-dependent localization of methane. Nuclear quantum effects and anharmonic sampling are both needed for a quantitative loaded-system C_P .
- The approximate combination

$$C \approx C_{\text{qn}}^{\text{har}} - C_{\text{cl}}^{\text{har}} + C_{\text{cl}}^{\text{anh}} \quad (68)$$

qualitatively recovers the loaded minimum and becomes useful above roughly 200 K in their 100-methane test.

The operational form used here is Eq. (91). Its classical MD term, its loaded harmonic frequencies, and its mass normalization are all evaluated for the same PET potential and the same loaded composition. The empty-framework Hessian is a reference calculation only. It cannot replace the loaded correction because it omits methane modes and host-guest coupling.

4 Present computational methodology

4.1 Structure preparation and methane insertion

The repository host structure is the conventional cubic MOF-5 cell with 424 atoms and a cell parameter of approximately 25.866 Å. The loaded campaign is generated by `python -m mof_heat_capacity.protocols.loaded`. For every temperature and replica, methane molecules are centered, uniformly randomly rotated using a quaternion, and placed at uniformly sampled fractional coordinates. Periodic atom-atom distances are checked against all previously placed atoms. A

100-methane structure contains 924 atoms. Independent insertion and velocity seeds prevent the nominal replicas from sharing an identical initial state.

This procedure prevents severe initial overlaps, but it is not an adsorption equilibrium calculation. It deliberately does not reproduce the RASPA insertion and Monte Carlo equilibration used by Kapil *et al.*; its adequacy must be shown by independent insertion seeds and sufficient subsequent equilibration before drawing conclusions about loading-dependent thermodynamics.

4.2 PET model and implementation

Section 2.2.4 describes the original PET architecture. The local PET-MAD and PET-SOL conversion metadata both specify the settings in Table 1. These describe the supplied checkpoints, not every PET model; identical architecture settings do not imply identical weights or training data. In particular, the original paper’s default of three attention layers per block differs from the one layer used here.

Table 1: Architecture settings recorded in the local PET-MAD and PET-SOL PET-JAX metadata. The species count is embedding capacity, not a measure of training coverage.

Setting	Value
Outer cutoff	8.0 Å
Cutoff switching width	0.5 Å
Adaptive neighbor-selection setting	40
Message-passing blocks	3
Attention layers per block	1
Token width, d_{PET}	256
Attention heads	8
Transformer feed-forward width	512
Node feature width, d_{node}	1024
Readout feature width, d_{head}	256
Species-embedding capacity	102

The locally inspected PyTorch implementation (`metatrain 2026.3.1`, `pet/modules/transformer.py`) embeds four geometric inputs, $(x_{ij}, y_{ij}, z_{ij}, r_{ij})$, rather than only the three Cartesian components in the original Eq. (50). Initial edge messages are species embeddings. Wider central features are contracted to the token width for attention and expanded afterward. With the dimensions in Table 1, each attention head has width $256/8 = 32$; `d_head=256` denotes a readout width. Padded neighbors are masked, and logarithms of cutoff factors are added to attention logits, implementing the weighting in Eq. (54) up to numerical regularization. In the residual feature path of `pet/model.py`, a reverse-neighbor index routes outgoing tokens to receiving environments; averaging the old message with the reversed new output updates the edge state. These are implementation details of the installed PyTorch package; the checkpoint version and PET-JAX conversion must remain compatible with them.

The model weights remain fixed at inference. The repository delegates the network implementation to the PET software stack: its export helper calls

`upet.save_upet` to turn the selected checkpoint into a portable metatomic model, which LAMMPS evaluates during production MD. Harmonic analysis converts the matching checkpoint into PET-JAX parameter and metadata files, then uses SADMOF to differentiate its scalar energy. The setup script pins PET-JAX to commit `ed0add4`; conversion transfers parameters rather than retraining a potential. Species baselines have zero Cartesian derivatives at fixed composition, whereas the learned energy scale must be retained in forces and Hessians.

Using matching checkpoints, exports, and conversions is essential to keep MD and harmonic analysis on a consistent PES. This must be checked by comparing energies and forces, including adaptive-cutoff derivative conventions, rather than inferred from matching architecture settings alone. The sparse Hessian path stops adaptive-cutoff derivatives, as described below. The selected metatomic export must expose stress, and campaign configurations record stress validation explicitly rather than inferring capability from a model filename. PET is the architecture; PET-MAD is a general-purpose fitted potential, whose energy, force, stress, and curvature accuracy for MOF-5 and its methane guests must still be validated [8].

4.3 Classical NPT molecular dynamics

The statistical and dynamical foundations of classical MD are summarized in Section 2.1. Only methane-loaded MOF-5 is propagated. LAMMPS evaluates a metatomic PET model in a flexible triclinic cell at an isotropic target pressure of 1 bar. Temperature and pressure are controlled by a Nosé–Hoover-chain/MTTK NPT fix, with chain length 3, thermostat damping time 100 fs, and barostat damping time 1000 fs. The production template specifies 10^6 steps of 0.5 fs, hence 500 ps, saves data every 100 steps (0.05 ps), and discards the first 100 ps. The initial campaign was subsequently extended to 175–425 K in 25 K increments for both PET-MAD and PET-SOL at 50, 100, and 150 methane molecules. The results reported here use that denser grid, with one replica at each state point. The 200–400 K hybrid reporting interval uses centered derivatives with neighboring points separated by 50 K. Independent replicas, longer production intervals, and temperature-spacing convergence remain to be established. The discarded interval is a configured choice, whose adequacy is assessed in Section 5.2.

The exported model must expose stress and must be validated for the intended MOF/loading states before NPT use. Mean loaded-system enthalpies are computed from the designated production portions of the trajectories,

$$\langle H \rangle_T = \langle E + P_{\text{ext}} V \rangle_T, \quad (69)$$

then averaged across replicas when available and differentiated with respect to temperature. This classical contribution retains guest diffusion, localization changes, and other anharmonic effects sampled by the MLIP.

4.4 Harmonic approximation and Cartesian Hessian

Let $\mathbf{R}_0$ be a stationary reference structure and $\mathbf{u} = \mathbf{R} - \mathbf{R}_0$. Expanding the potential energy gives

$$U(\mathbf{R}_0 + \mathbf{u}) = U(\mathbf{R}_0) + \underbrace{\nabla U(\mathbf{R}_0) \cdot \mathbf{u}}_{=0 \text{ at a minimum}} + \frac{1}{2} \mathbf{u}^T \mathbf{H} \mathbf{u} + \mathcal{O}(u^3), \quad (70)$$

with Cartesian Hessian

$$H_{i\alpha,j\beta} = \frac{\partial^2 U}{\partial R_{i\alpha} \partial R_{j\beta}} = -\frac{\partial F_{i\alpha}}{\partial R_{j\beta}}. \quad (71)$$

For N atoms, $\mathbf{H}$ has dimensions $3N \times 3N$; for the 100-methane cell this is 2772×2772 .

4.4.1 Automatic differentiation through a neural network

A neural-network potential is a composition of elementary differentiable operations. Schematically, with the model parameters θ fixed during inference,

$$\mathbf{z}^{(0)} = \mathbf{R}, \quad \mathbf{z}^{(\ell+1)} = \phi_\ell(\mathbf{z}^{(\ell)}; \theta), \quad qquad U_\theta = \rho(\mathbf{z}^{(L)}). \quad (72)$$

The functions ϕ_ℓ include PET’s coordinate encoding, attention, message updates, cutoff weighting, and non-linear layers, while ρ is the scalar energy readout. Automatic differentiation (AD) records this computational graph and applies the chain rule to the derivative rules of its elementary operations [16]. It therefore differentiates the energy function implemented by the model, up to floating-point arithmetic, without choosing the displacement step required by finite differences. AD does not improve the learned potential itself: an exact derivative of an inaccurate U_θ remains an inaccurate force or curvature.

It is useful first to separate a derivative *operator* from its matrix representation. For a differentiable map $\mathbf{f} : \mathbb{R}^n \rightarrow \mathbb{R}^m$, the Jacobian is

$$(\mathbf{J}_f)_{pq} = \frac{\partial f_p}{\partial x_q}, \quad \mathbf{J}_f \in \mathbb{R}^{m \times n}. \quad (73)$$

Forward-mode AD evaluates the Jacobian–vector product (JVP) $\mathbf{J}_f \mathbf{v}$ without constructing $\mathbf{J}_f$, whereas reverse-mode AD evaluates the vector–Jacobian product (VJP) $\mathbf{w}^T \mathbf{J}_f$. Consequently, a dense Jacobian can be materialized column by column using n JVPs with the coordinate vectors $\mathbf{e}_q$, or row by row using m VJPs. The intermediate Jacobians of the neural-network layers need never be stored: AD propagates only tangent values in forward mode or adjoint values in reverse mode [17].

Reverse-mode AD propagates one scalar output backward through the graph and is well suited to obtaining all $3N$ components of $\mathbf{g} = \nabla_{\mathbf{R}} U_\theta$ in one reverse sweep. The model forces are $\mathbf{F} = -\mathbf{g}$. Differentiating this gradient again would give the full Hessian. Rather than materializing it directly, PET-JAX and SADMOF use

a forward-mode directional derivative of the reverse-mode gradient, commonly called forward-over-reverse AD. For a seed direction $\mathbf{v}$ this evaluates

$$\mathbf{H}\mathbf{v} = \left. \frac{d}{d\epsilon} \nabla_{\mathbf{R}} U_{\theta}(\mathbf{R} + \epsilon \mathbf{v}) \right|_{\epsilon=0}. \quad (74)$$

Equation (74) is a Hessian–vector product (HVP). Using each Cartesian basis vector in turn would recover all Hessian columns, but would still require $3N$ differentiated evaluations. SADMOF reduces that count by exploiting locality.

The validity of these derivatives also depends on the differentiability of the complete model. In particular, the cutoff factors illustrated in Figures 1 and 2 occur inside the computational graph. Their smoothness controls the continuity of forces and Hessians as neighbors approach the cutoff. Discrete neighbor selection is a separate issue and is handled by the fixed-graph derivative convention described below.

4.4.2 Sparse Hessian reconstruction with SADMOF

For a local message-passing potential, the energy associated with an atom can depend only on atoms reachable through a finite number of neighbor-graph hops. Consequently, many atom-pair blocks $\mathbf{H}_{ij} \in \mathbb{R}^{3 \times 3}$ are structurally zero. SADMOF [18] first uses PET’s selected-neighbor graph to construct a Boolean sparsity pattern $\mathbf{S}$ at a specified reachability depth $\mathbf{hops}$, then expands every connected atom pair into its Cartesian 3×3 block.

The supplied paper by Langer, Hill, and Ceriotti [18] derives this reach from the readout architecture (Section 4 and Appendix C). With L message-passing layers, a node-readout energy term couples atoms within one L -hop ball, giving maximum Hessian reach $K = 2L$. PET’s edge readout couples the L -hop neighborhoods of two adjacent atoms, giving $K = 2L + 1$. Thus the paper’s MACE-MP-0 medium, PET-XS, and PET-S models have exact reaches of four, five, and seven hops, respectively. If $\mathbf{A}$ is the undirected input-graph adjacency, the atom-pair pattern is

$$\mathbf{P}^{(k)} = \mathbf{I} \vee \mathbf{A} \vee \mathbf{A}^2 \vee \dots \vee \mathbf{A}^k, \quad (75)$$

with powers and unions evaluated in Boolean arithmetic. At $k = K$ this includes every structurally allowed coupling under the stated derivative convention; $k < K$ defines the paper’s truncated ASD hierarchy. Additional nonlocal energy terms would require a separate analysis.

Here “structurally zero” is stronger than being numerically zero at one geometry. Writing Cartesian indices as $p, q \in \{1, \dots, 3N\}$, the pattern is intended to satisfy, for every geometry covered by the pattern,

$$S_{pq} = 0 \implies H_{pq} = 0 \quad (76)$$

An entry allowed by $S_{pq} = 1$ may nevertheless vanish at a particular geometry because of symmetry or cancellation. Such an overestimate merely stores extra entries and can require extra colors; an underestimate with $S_{pq} = 0$ where $H_{pq} \neq 0$ discards curvature and

can also contaminate retained entries during decomposition. For a Hessian, detecting the pattern is intrinsically a second-order problem: linear dependence can later participate in a nonlinear operation and create a mixed second derivative. The general automatic sparse differentiation pipeline therefore tracks nonlinear and cross-argument dependence, while SADMOF obtains a conservative pattern from the finite receptive field of the local interatomic potential [17].

The simplest compression scheme ignores Hessian symmetry and forms the column-intersection graph $\mathcal{G}_S$. Its vertices are the Cartesian columns, and two distinct columns are adjacent precisely when their possible supports overlap:

$$(q, r) \in \mathcal{E}_S \iff \exists p : S_{pq} = S_{pr} = 1. \quad (77)$$

A proper coloring $c(q) \in \{1, \dots, n_c\}$ assigns different colors to adjacent columns. Columns of one color are then *structurally orthogonal*: no output row can contain a possible nonzero from two of them. Finding the minimum number of colors is NP-hard in general, so practical ASD software uses inexpensive greedy heuristics. Optimality affects speed, not the reconstructed values: any valid coloring is sufficient [17].

For this nonsymmetric column-coloring construction, let $\mathcal{C}_a$ be one color class and define the compressed seed

$$\mathbf{v}_a = \sum_{q \in \mathcal{C}_a} \mathbf{e}_q, \quad \mathbf{H}\mathbf{v}_a = \sum_{q \in \mathcal{C}_a} \mathbf{H}\mathbf{e}_q. \quad (78)$$

Collecting these seeds as columns of $\mathbf{V} \in \{0, 1\}^{3N \times n_c}$, with $V_{qa} = 1$ exactly when $c(q) = a$, gives the compressed derivative matrix

$$\mathbf{Y} = \mathbf{H}\mathbf{V}, \quad Y_{pa} = \sum_{q: c(q)=a} H_{pq}. \quad (79)$$

The pattern makes the contributing column unambiguous in every row. In fact, for every entry permitted by the pattern, direct decompression is simply

$$H_{pq} = Y_{p, c(q)} \quad \text{when } S_{pq} = 1, \quad (80)$$

and all entries with $S_{pq} = 0$ are set to zero. Equation (80) works because a proper coloring guarantees that every other term in the sum in Eq. (79) is zero in row p .

SADMOF’s ASDEX Hessian coloring instead uses the Hessian symmetry $H_{pq} = H_{qp}$ through a star coloring. A star coloring is a proper coloring of the undirected Hessian sparsity graph in which every path on four vertices uses at least three colors; equivalently, each subgraph induced by any two colors is a collection of stars. This weaker condition allows some column-support collisions that Eq. (80) alone could not decode. For each unordered nonzero pair $\{p, q\}$, the coloring records a hub–spoke orientation for which one of the two compressed locations is uncontaminated. ASDEX gathers that value and writes it to both H_{pq} and H_{qp} . Symmetry thus acts as a second possible route to each off-diagonal entry and can reduce n_c relative to ordinary column coloring [17, 18].

Appendix D of the paper [18] also exploits the 3×3 block structure: star-color the atom-pair graph, then assign Cartesian component $d \in \{0, 1, 2\}$ of atom i the color $3c_i + d$. This lifts the atom coloring to a valid coordinate star coloring while reducing setup cost. The present wrapper passes the pattern directly to AS-DEX; availability of this optimization depends on the installed dependency implementation.

The complete compression pipeline is therefore: detect $\mathbf{S}$, color the associated graph, evaluate the n_c colored HVPs, and decompress their nonzero entries. Dense forward-over-reverse materialization needs $3N$ HVPs; the sparse route needs n_c , so its expensive differentiated work is reduced by an ideal factor of approximately $3N/n_c$. Pattern construction, coloring, decomposition, compilation, and GPU utilization add overhead, so this ratio is not a wall-clock guarantee. The setup cost can be amortized over repeated Hessians only while the structural pattern remains valid. Several colors may also be evaluated together as one GPU batch; batching changes throughput and memory use but not the number of mathematical seed directions.

The project implementation follows six steps. It converts the ASE structure to padded PET-JAX positions, cell, species, masks, and graphs with `atoms_to_inputs`; constructs the pattern with `sparsity_pattern`; obtains a forward-over-reverse coloring with `asdex.hessian_coloring_from_sparsity`; constructs the scalar PET energy with `get_energy_fn`; builds the colored HVP evaluator with `get_hessian_fn`; and compiles it for the GPU using `jax.jit`. The returned JAX sparse array is finally densified, PET-JAX padding atoms are removed, and the real-atom $(3N) \times (3N)$ matrix is passed to the normal-mode calculation. Sparsity therefore reduces the expensive AD work; the present eigendecomposition still consumes a dense matrix.

The current calls use `get_energy_fn(..., no_shadow=True)`. This stops derivatives through PET’s adaptive neighbor-selection mechanism and differentiates the energy on the selected graph. Allowing shadow derivatives introduces couplings outside that sparse graph and requires a dense treatment, so the project rejects `shadow=true` in this path. This convention must be checked for consistency with the forces used in MD.

Sparse reconstruction is not a modified physical definition of the Hessian. It reproduces the dense AD Hessian only when $\mathbf{S}$ contains every nonzero entry of the differentiated model. In particular, `hops` is the reach of the proposed Hessian pattern, not the number of PET message-passing blocks. The paper derives seven hops for three-layer PET-S ($K = 2L + 1$), using the selected graph and omitting adaptive-cutoff forces [18]. The project’s current value `hops=3` is therefore a truncation and must be increased until the frequencies and heat-capacity correction converge.

Crucially, truncated ASD is not equivalent to zeroing distant blocks of an already computed dense Hessian. Its HVPs still differentiate the complete energy callable; couplings omitted from the assumed pattern may share colors with retained entries and contaminate

their compressed values. With $\mathbf{M}_k$ the Cartesian mask and $\widetilde{\mathbf{H}}_k$ the reconstructed Hessian before acoustic-sum-rule postprocessing, write

$$\Delta \mathbf{H}_{\text{disc}} = -(\mathbf{1} - \mathbf{M}_k) \odot \mathbf{H}, \quad (81)$$

$$\Delta \mathbf{H}_{\text{cont}} = \mathbf{M}_k \odot (\widetilde{\mathbf{H}}_k - \mathbf{H}), \quad (82)$$

where $\odot$ denotes entrywise multiplication. The supports are disjoint, so

$$\|\widetilde{\mathbf{H}}_k - \mathbf{H}\|_F^2 = \|\Delta \mathbf{H}_{\text{disc}}\|_F^2 + \|\Delta \mathbf{H}_{\text{cont}}\|_F^2. \quad (83)$$

Appendix H finds that contamination tracks the discarded norm until the floating-point floor; this is empirical evidence, not a universal error bound. Acoustic-sum-rule enforcement does not undo either general source of error.

The paper reports rapid decay of force constants with hop distance and only modest speedups for exact patterns. At a 0.1% heat-capacity tolerance at 300 K, most benchmark structures converge at two hops for MACE, three for PET-S, and four for PET-XS, with slower-converging zeolite cases [18]. Those observations do not establish convergence for loaded MOF-5. Compare increasing hop counts against an exact-pattern or feasible dense reference at fixed geometry and derivative convention, inspecting signed frequencies and the final harmonic correction across the temperature grid. Small spectral shifts near a mode-selection threshold can change heat capacity, and convergence need not be monotonic (Appendix E).

Exactness at a fixed cell size is also distinct from periodic-cell convergence. Appendix A tracks periodic image offsets through the model’s full graph reach to choose supercells without folding distinct interactions onto one force-constant entry. Sparse AD alone does not establish that a chosen cell provides a converged vibrational heat capacity.

The other numerical controls do not change the intended Hessian: `chunk_size` controls how many colored directions are batched, `remat=true` asks JAX to recompute selected intermediates during the backward pass to reduce memory, and `dtype` selects floating-point precision. The configured calculation uses `float32`, `chunk_size=1`, and rematerialization. These settings must be checked against higher precision and alternative resource settings; changing `hops`, by contrast, can change the mathematical sparsity approximation itself.

None of the resource settings is presently an established convergence value. Comparisons must use separately tagged outputs and should include a dense or less aggressively truncated reference where feasible, especially for the low-frequency modes.

4.5 Normal modes and quantum harmonic heat capacity

The Hessian is symmetrized and mass weighted to obtain

$$D_{i\alpha,j\beta} = \frac{H_{i\alpha,j\beta}}{\sqrt{m_i m_j}}, \quad D e_k = \omega_k^2 e_k. \quad (84)$$

The code enforces the force-constant acoustic sum rule before diagonalization,

$$\sum_j H_{i\alpha,j\beta} = 0, \quad (85)$$

so that uniform translations have zero restoring force. The upstream SADMOF observable maps negative eigenvalues to zero frequency for compatibility with its reference heat-capacity convention. The present workflow instead diagonalizes the mass-weighted Hessian locally and retains unstable eigenvalues as signed negative frequencies. Hybrid assembly rejects unexpected imaginary modes and permits at most the three translational near-zero modes in its strict mode. The available results instead use an explicitly exploratory option that discards modes at or below the frequency threshold; their stability failures are documented in Section 5.5.

For each retained mode, define

$$x_k = \frac{\hbar\omega_k}{k_B T} = \frac{\hbar c \tilde{\nu}_k}{k_B T}, \quad (86)$$

where $\tilde{\nu}_k$ is the wavenumber. The quantum harmonic contribution is

$$C_{V,k} = k_B \left[\frac{x_k/2}{\sinh(x_k/2)} \right]^2, \quad (87)$$

and the gravimetric result is

$$c_V(T) = \frac{1}{M_{\text{cell}}} \sum_{k \in \mathcal{V}} C_{V,k}, \quad (88)$$

in J/g/K when M_{cell} is the mass of one cell in grams and $C_{V,k}$ is in J/K. Equivalently, one uses N_A/M_{molar} for the molar mass of that cell composition. The mass includes methane for a loaded system, so both the spectrum and the normalization change with loading.

One Hessian produces the entire configured 100–500 K curve in 10 K increments. The temperature appearing in Eq. (88) controls oscillator populations; it does not require a separate MD trajectory at every analysis temperature when the force constants are assumed temperature independent.

4.6 Optimized minima and the hybrid correction

The empty-framework calculation starts from the equilibrated MOF-5 structure already available to the project. Its coordinates are relaxed at fixed cell with the selected MLIP and one AD Hessian is evaluated. No empty-MOF MD is needed merely to obtain a Hessian reference.

The target protocol quenches final loaded structures from several independent classical MD replicas at fixed cell with the same MLIP. The current outputs use one reference structure per model and loading, taken at 400 K; Section 5.5 distinguishes their exploratory mode handling from the acceptance criteria below. In the strict protocol, a Hessian is accepted only after the relaxation reaches the chosen force tolerance and its signed spectrum is checked for unexpected imaginary

modes. Raw finite-temperature frames are not treated as normal-mode reference structures. If N_m independent optimized minima are retained, the mean harmonic quantum correction is

$$\overline{\Delta c^{\text{har}}}(T) = \frac{1}{N_m} \sum_{s=1}^{N_m} \left[c_{\text{qn},s}^{\text{har}}(T) - c_{\text{cl},s}^{\text{har}} \right], \quad (89)$$

with sample variance

$$s_{\Delta c}^2(T) = \frac{1}{N_m - 1} \sum_{s=1}^{N_m} \left[\Delta c_s^{\text{har}}(T) - \overline{\Delta c^{\text{har}}}(T) \right]^2. \quad (90)$$

For each minimum, the quantum and classical harmonic terms use exactly the same retained-mode mask. The classical term is k_B per retained mode. The complete loaded-system spectrum is used because framework and methane motion can mix; the empty Hessian alone omits methane quantization and cannot supply this correction.

The final estimator is

$$c_P^{\text{approx}}(T) = c_{P,\text{cl}}^{\text{anh}}(T) + \overline{\Delta c^{\text{har}}}(T), \quad (91)$$

$$c_{P,\text{cl}}^{\text{anh}}(T) = \frac{1}{M_{\text{cell}}} \frac{d\langle H \rangle_{\text{cl}}}{dT}.$$

Extensive terms are combined before division by the total mass of the loaded cell. Replica-aware resampling propagates uncertainty in the enthalpy curve; the spread across optimized minima measures sensitivity of the harmonic correction. The Hessian contribution is formally a C_V correction added to a classical C_P term, following the small solid-state $C_P - C_V$ assumption used in the reference approximation. This mixing and the approximation’s limited validation below roughly 200 K must be stated with the result.

Persistent LLPR members contribute both a classical term and a harmonic correction. Their model uncertainty must therefore be combined before taking a standard deviation:

$$\sigma_{\text{model}}^2 = \sigma_{\text{cl,model}}^2 + \sigma_{\text{har,model}}^2 + 2 \text{cov}_i(c_{P,\text{cl}}^{(i)}, \Delta c_{\text{har}}^{(i)}), \quad (92)$$

$$u_{\text{hyb}}^2 = \sigma_{\text{MD}}^2 + \sigma_{\text{har,samp}}^2 + \sigma_{\text{model}}^2. \quad (93)$$

All terms use mass-specific units. The covariance is across the same model members, not across trajectory frames. Here σ_{MD} is the sampling standard error of the central enthalpy derivative and $\sigma_{\text{har,samp}}$ is the error across independently quenched reference structures. The latter is distinct from the harmonic LLPR spread. Equation (93) assumes that the sampling and model contributions being added in quadrature are independent.

For the available single-replica calculations, enthalpy means are perturbed with Gaussian draws using their autocorrelation-based standard errors and the complete temperature derivative is recomputed. This propagates estimated within-trajectory uncertainty, but is

not a block bootstrap of the raw trajectories and supplies no between-replica variance. The plotted bands use $u_{\text{hyb}} = (\sigma_{\text{MD}}^2 + \sigma_{\text{model}}^2)^{1/2}$: there is only one quenched geometry per model and loading, so harmonic sampling uncertainty is unestimated and is excluded rather than established to be zero. The member spread is evaluated at central-model geometries and does not include member-specific geometry relaxation.

5 Results and discussion

5.1 Available calculations and status of the estimates

The available post-processing covers two checkpoints, PET-MAD (`pet-mad-1.5-s-40nn`) and PET-SOL (`pet-sol-s-best`), at 50, 100, and 150 methane molecules per conventional MOF-5 cell. These compositions contain 674, 924, and 1174 atoms, respectively. Each of the six model-loading combinations has one replica at eleven temperatures from 175 K to 425 K, spaced by 25 K: 66 trajectories in total. Each trajectory spans 500 ps; the analysis discards 100 ps and uses 8001 saved frames over the remaining 400 ps. This gives 33 ns of simulated time and 26.4 ns of analyzed production time across the campaign, rather than independent replicas at any one state point.

The hybrid curves cover nine temperatures from 200 to 400 K. The additional MD points at 175 and 425 K provide the neighboring enthalpies needed for centered derivatives at the ends of this reporting interval. The central classical term is the derivative of the trajectory-driving potential’s physical mean enthalpy, as in Eq. (91); it is not the mean of the reweighted LLPR committee. Each loaded composition uses one fixed-cell quenched structure taken from its 400 K trajectory. The same spectrum is evaluated at every reported temperature. There are therefore six loaded reference geometries, not nine independent geometries per curve.

The results below are *exploratory hybrid estimates*. The archived calculations deliberately discard imaginary and near-zero modes, and four of the six loaded relaxations did not satisfy their requested convergence criterion. Even the two force-converged structures have substantial imaginary spectra. These outputs therefore do not meet the stable-minimum acceptance conditions of Section 4.6. They nevertheless reveal the size of the classical and quantum terms, loading trends, and the limitations of the present uncertainty propagation. The plotted bands quantify selected sampling and model contributions, not the error caused by unstable reference structures or incomplete exploration of configuration space.

Numerical trajectory diagnostics are taken from the per-run `summary.json` files in the trajectory-analysis directory. Central hybrid values and relaxation diagnostics come from the assembled `heat-capacity.exploratory-discard-imaginary` CSV and JSON files. The uncertainty tables accompanying the loading-comparison figures are used for the

reported CEA and Gaussian bands: they incorporate the refreshed stride-1 evaluation of all 8001 frames with 64 persistent LLPR members. Earlier uncertainty columns in the per-loading hybrid CSVs are not substituted for these refreshed values. Figure snapshots and their source paths are recorded in the report’s `results-provenance.json`.

5.2 Thermodynamic sampling and equilibration

The thermostat maintains the mean temperature close to its target throughout the campaign: the largest absolute deviation is 0.315 K. For example, PET-MAD with 100 methane molecules at 300 K gives $\langle T_{\text{kin}} \rangle = 299.854$ K and a temperature standard deviation of 8.031 K. Equipartition with $f = 3N - 3 = 2769$ predicts $T\sqrt{2/f} = 8.058$ K, in close agreement with the observed width. This is a useful check of the kinetic distribution discussed in Section 2.1, although it tests neither configurational ergodicity nor force-field accuracy.

Instantaneous pressure has a standard deviation of approximately $0.80\text{--}1.51 \times 10^3$ bar across the runs, whereas standard errors of the mean are approximately 9–18 bar. Such fluctuations are consistent with the large kinetic and virial contributions in a small periodic cell [Eq. (16)]. For the same PET-MAD 100-methane example, $\langle P_{\text{inst}} \rangle = 4.65 \pm 13.11$ bar, compatible with the 1 bar target. Pressure noise must not be inserted into the enthalpy: the heat-capacity calculation uses $H = E + P_{\text{ext}}V$ with fixed P_{ext} . Thermostat/barostat control is consequently supported by these diagnostics, but is not by itself an equilibration criterion.

The enthalpy curves increase smoothly with temperature (Figure 6), giving positive classical heat capacities. The archived enthalpy autocorrelation times are 0.095–0.101 ps, corresponding to approximately 1980–2115 effective samples per trajectory instead of 8001 independent observations. Their mean enthalpy standard errors are 0.015–0.046 eV. The distinction between saved and independent samples is exactly the one in Eq. (37). However, the short estimated correlation time describes the fluctuations resolved by this estimator; it does not exclude slower drift or unvisited states.

Indeed, 33 of the 66 runs have an absolute enthalpy split statistic greater than two. Here the stored diagnostic is

$$z_H = \frac{\overline{H}_{\text{second}} - \overline{H}_{\text{first}}}{\sqrt{\text{SE}_{\text{first}}^2 + \text{SE}_{\text{second}}^2}}. \quad (94)$$

For PET-MAD with 150 methane molecules at 175 K, $z_H = 7.47$; for PET-SOL with 150 molecules at 300 K, it is 4.57. A threshold crossing is a diagnostic rather than a formal independent hypothesis test, but the frequency and size of these discrepancies preclude a blanket claim that 100 ps of equilibration was sufficient. Some RMS-force time series have correlation times as long as 26.1 ps, with fewer than eight estimated effective samples, illustrating that convergence is observable dependent.

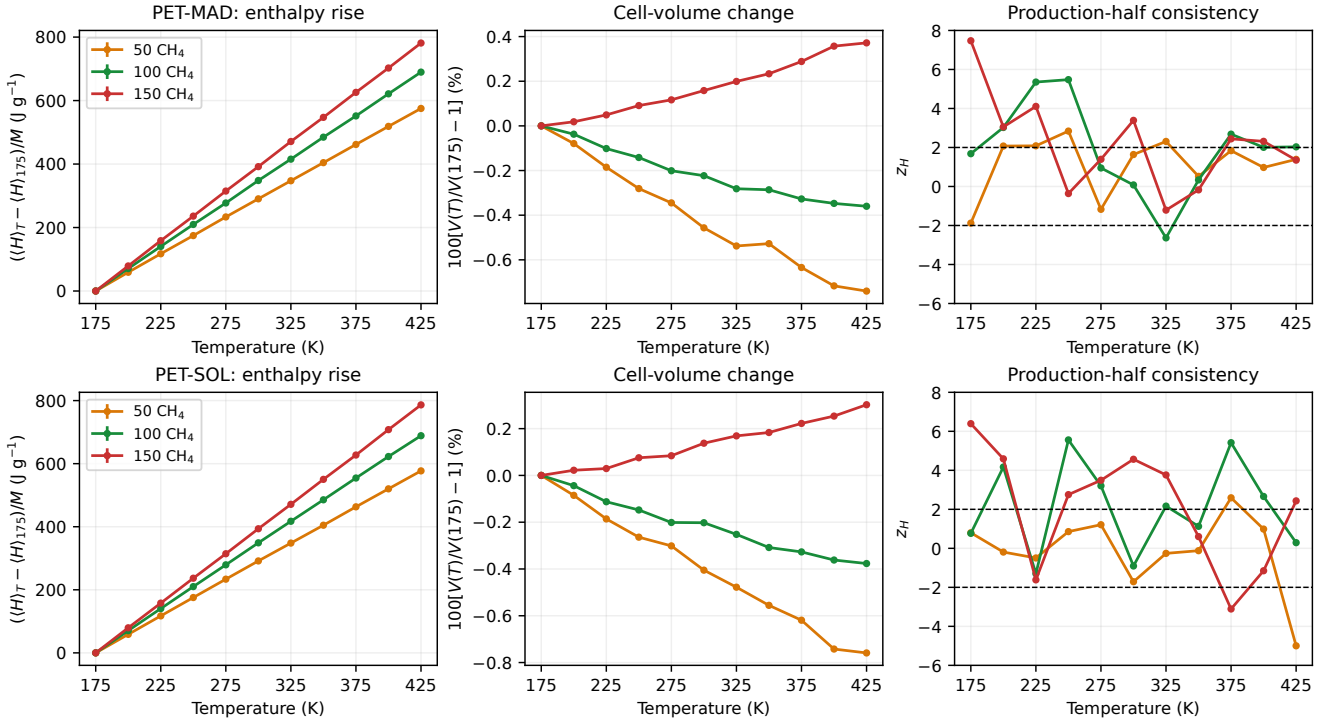


Figure 6: Thermodynamic diagnostics from all 66 trajectory summaries. Left: mass-specific enthalpy increase relative to the same model and loading at 175 K; the baseline is a display offset, and the small bars show the pointwise enthalpy standard errors without propagating the shared baseline error. Middle: fractional change of mean cell volume relative to 175 K. Right: the archived difference between production-half mean enthalpies divided by their combined standard error, z_H ; dashed lines mark ± 2 . These are diagnostics of the individual runs, not replica averages.

Figure 7 makes the issue concrete: in the 175 K PET-MAD example the accumulated mean still changes appreciably as production is extended. Its offset is about -0.30 eV after 100 ps and about -0.15 eV after 200 ps, relative to the final mean. The much larger LLPR enthalpy spread in the upper panel does not make this drift harmless: heat capacity differentiates enthalpy, and a persistent energy offset across temperatures can cancel whereas a temperature-dependent sampling bias does not. Longer runs, analysis with alternative discarded intervals, and independently initialized replicas are needed before interpreting the small reported MD errors as complete sampling uncertainties.

5.3 Cell response, framework motion, and methane mobility

The volume response changes with methane loading. Between 175 and 425 K, PET-MAD predicts a decrease from 17146.8 to 17019.9 $\text{\AA}^3$ at 50 molecules, a smaller decrease from 17170.2 to 17108.4 $\text{\AA}^3$ at 100 molecules, and an increase from 17210.4 to 17274.4 $\text{\AA}^3$ at 150 molecules. PET-SOL shows the same qualitative pattern. The corresponding endpoint volume changes are about -0.75% , -0.37% , and $+0.3$ – 0.4% across the two models. This suggests that guest loading modifies the framework’s thermal response, with contraction at lower loading and expansion at the highest loading. The mean densities at 300 K are approximately 0.677, 0.752, and 0.825 g/cm 3 for 50, 100, and 150 molecules,

respectively; most of this loading trend reflects the added methane mass.

This observation connects directly to Eq. (31): thermal expansion enters $C_P - C_V$ through α_V^2 , so negative thermal expansion does not imply a negative $C_P - C_V$. The volume trend alone neither establishes a phase transition nor quantifies the neglected quantum pressure–volume correction; compressibility and consistently converged derivatives would also be needed.

At 300 K, the six mean framework RMSDs lie between 0.659 and 0.725 $\text{\AA}$, while the RMS changes of the tracked reference-bond lengths are only 0.055–0.059 $\text{\AA}$. In the PET-SOL 150-molecule example (Figure 8), the framework fluctuates around a bounded RMSD instead of displaying continued displacement growth. This combination is consistent with collective framework motion without a gross collapse of the tracked structure over the analyzed interval. It is not a complete test of chemical integrity: the diagnostics track a reference neighbor set and do not establish the absence of every possible bond change.

Methane remains mobile. At 300 K, the 400 ps MSD is approximately 8.9 – $9.2 \times 10^3 \text{\AA}^2$ at 50 molecules, 5.1 – $5.6 \times 10^3 \text{\AA}^2$ at 100 molecules, and 3.1 – $3.4 \times 10^3 \text{\AA}^2$ at 150 molecules. The lower displacement at higher loading is consistent with hindered guest motion as the pores become more crowded. The methane center-of-mass radial distributions have their first prominent peak near 4 $\text{\AA}$, showing local guest organization alongside translational mobility. These diagnostics support

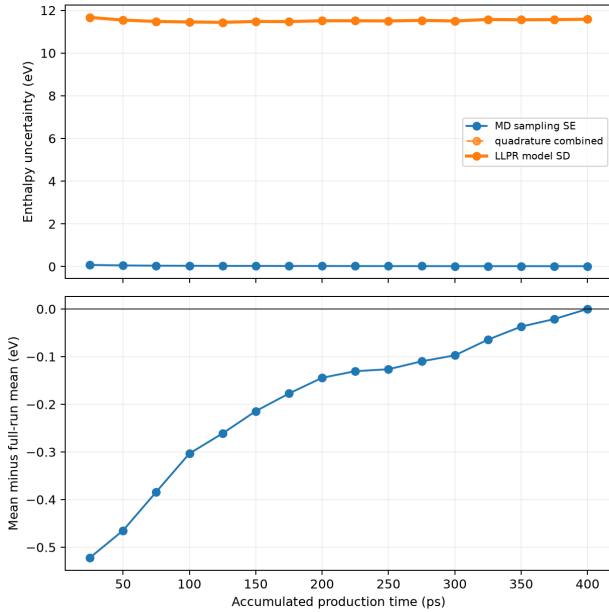


Figure 7: Archived enthalpy convergence diagnostic for PET-MAD with 150 methane molecules at 175 K. The upper panel separates MD sampling error from the LLPR spread of the *enthalpy*, in eV. The lower panel shows the accumulated mean relative to the full-production mean. Its endpoint is zero by construction and is not evidence of equilibration.

the need for the anharmonic MD term in Eq. (91): diffusion and changing adsorption environments cannot be generated by a harmonic expansion about one geometry. No diffusion coefficient is inferred here from the endpoint MSD; a defensible fit would require a verified diffusive regime, control of cell-motion effects, adequate time-origin statistics, and replica uncertainty.

5.4 Classical, harmonic, and hybrid heat capacities

Figures 9 and 10 show the supplied loading-comparison curves with the refreshed CEA uncertainty bands. Table 2 resolves the contributions at 300 K. For PET-MAD, increasing loading from 50 to 150 molecules raises the classical term from 2.281 to 3.122 J/g/K; for PET-SOL the corresponding range is 2.289–3.134 J/g/K. The quantum-minus-classical harmonic correction is negative, ranging from approximately -1.35 to $-1.97 \text{ J g}^{-1} \text{ K}^{-1}$ across the six systems. It removes roughly 59–63% of the classical heat capacity at this temperature.

The large reduction is the expected signature of classical equipartition populating high-frequency framework and methane vibrations too strongly. It follows from the quantum oscillator contribution in Eq. (87): a thermally inaccessible mode contributes less than the classical k_B . The increasingly negative correction with loading also demonstrates why the empty-framework spectrum cannot replace the loaded-system Hessian. Both the additional methane internal modes and mixed host-guest modes enter the correction.

The net room-temperature hybrid values are approximately 0.93, 1.07–1.08, and 1.16 J/g/K for 50, 100, and 150 methane molecules. The PET-MAD and PET-SOL central values differ by only 0.003, 0.011, and 0.001 J/g/K, respectively. These differences are much smaller than the propagated model bands, so the calculations do not distinguish the two potentials’ predictive accuracy. Agreement can also reflect shared modeling assumptions and cannot be used as independent validation of the harmonic approximation.

Across 200–400 K, the PET-MAD loaded curves rise from 0.796, 0.932, and 1.014 to 1.106, 1.278, and 1.362 J/g/K. PET-SOL gives corresponding endpoints 0.787, 0.921, and 0.987, and 1.102, 1.189, and 1.422 J/g/K. Increasing thermal occupation makes the harmonic correction less negative, contributing to the upward temperature trend even where the classical term changes little. For example, the PET-MAD 50-molecule classical term changes from 2.347 to 2.269 J/g/K, while its correction changes from -1.551 to $-1.164 \text{ J g}^{-1} \text{ K}^{-1}$. The positive slope of the final curve therefore should not be attributed solely to increasing classical anharmonicity.

The broad ordering with loading is common to both models, but adjacent loading bands overlap substantially. Quantitative significance of a loading difference would require propagating the *difference* with persistent member identities, not merely inspecting overlap of marginal bands. The reported range starts at 200 K; it does not bracket a low-temperature minimum and cannot verify the minimum discussed in Section 3.2. Small local bends, such as the PET-SOL 100-molecule turnover near 400 K, remain sensitive to finite sampling and temperature differentiation.

The empty-system values at 300 K are 0.786 ± 0.021 for PET-MAD and $0.766 \pm 0.027 \text{ J g}^{-1} \text{ K}^{-1}$ for PET-SOL. These are quantum-harmonic c_V values with LLPR model standard deviations, separately normalized by the framework mass. They are not a full hybrid c_P and do not measure the incremental heat capacity of adsorbing one methane molecule. The dashed reference curves were digitized from Kapil *et al.* [6], as confirmed for the supplied reference CSVs. At 300 K those files give approximately 0.788, 0.944, 1.037, and 1.151 J/g/K for 0, 50, 100, and 150 methane molecules. The broad loading and temperature trends agree with the present curves, and the loaded room-temperature differences are at most about 0.043 J/g/K. This is a contextual comparison, not validation: the digitized points contain no uncertainty estimates, the underlying force fields and preparation protocols differ (Section 3.1), and the present spectra and sampling have unresolved limitations. The approximate empty-framework scale quoted earlier is less precise than these digitized values and should not be treated as a separate numerical target.

5.5 Hessian stability and interpretation of the quantum correction

Table 3 prevents interpreting a small residual force as proof of a valid harmonic minimum. The two 100-

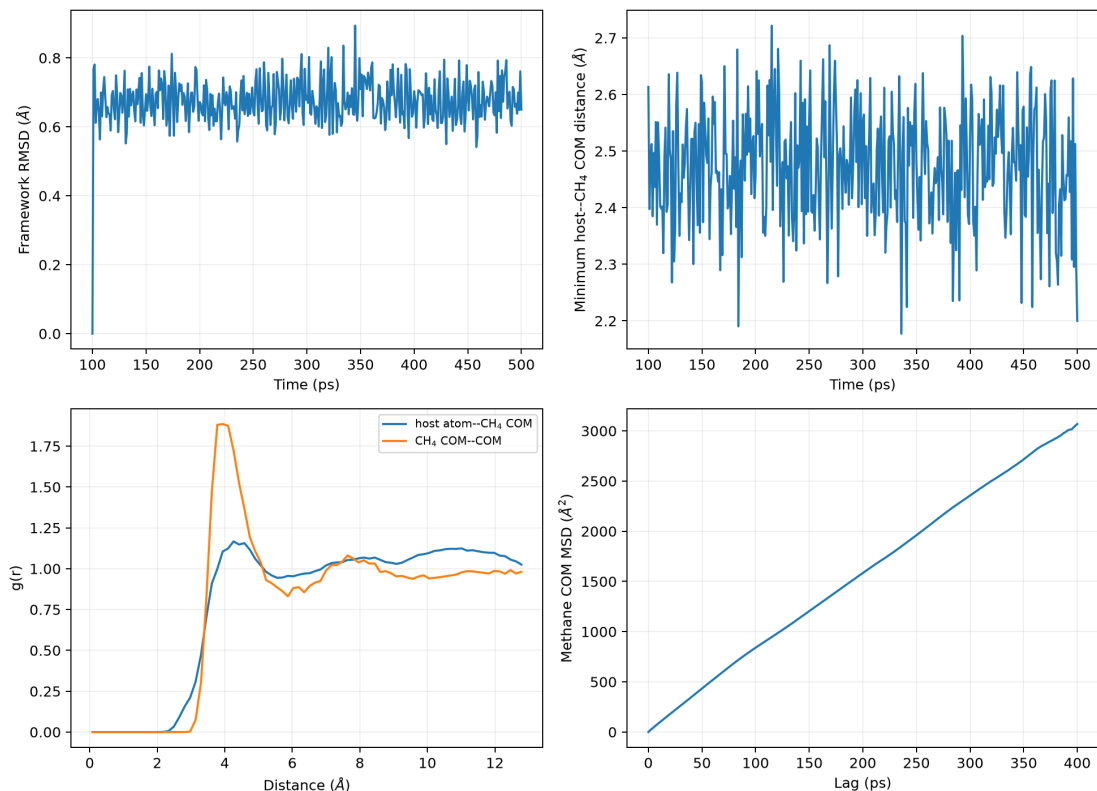


Figure 8: Structural diagnostics for PET-SOL with 150 methane molecules at 300 K: framework RMSD, nearest host-atom-methane center-of-mass distance, radial distributions, and methane center-of-mass mean-squared displacement (MSD). The first analyzed frame at 100 ps defines the framework reference, hence the initial zero RMSD. Long-lag MSD values use fewer time origins than short-lag values.

molecule structures meet their archived force criteria, yet have 114 and 95 imaginary modes for PET-MAD and PET-SOL. The most negative signed frequencies are -92.4 and -138.4 cm^{-1} , far beyond the 1 cm^{-1} exclusion threshold. The other loadings retain 17–35 imaginary modes and are flagged unconverged. The force tolerances are run dependent: a smaller final force in one run does not override that run’s failed convergence flag. PET-MAD at 50 molecules also has five near-zero modes, exceeding the three translational modes expected for a stable periodic structure.

Section 4.5 relates squared frequencies to eigenvalues of the mass-weighted Hessian. Negative eigenvalues signal directions in which the quadratic energy decreases. Discarding them makes the oscillator sum numerically evaluable, but does not make the reference geometry a stable minimum. The cause cannot be assigned from these archives alone: insufficient relaxation, unstable stationary structures, sparse-Hessian truncation, or other curvature errors must be distinguished by convergence checks. The empty spectra also contain four and five imaginary modes, with minima of -5.5 and -6.6 cm^{-1} , so their stability is not established either, although their negative curvatures are much smaller.

Using the same retained-mode mask in the quantum and classical terms avoids subtracting a different number of modes, as required by Section 4.6. It does not quantify the missing anharmonic physics of an unstable direction. Nor does reusing one 400 K

spectrum establish that its curvature represents every temperature. With one quenched geometry per model and loading, the stored zero harmonic sampling error means that minimum-to-minimum variation was not estimated. The LLPR Hessian spread varies the energy readout at the fixed central geometry; it does not replace that missing structural sampling or include member-specific relaxation.

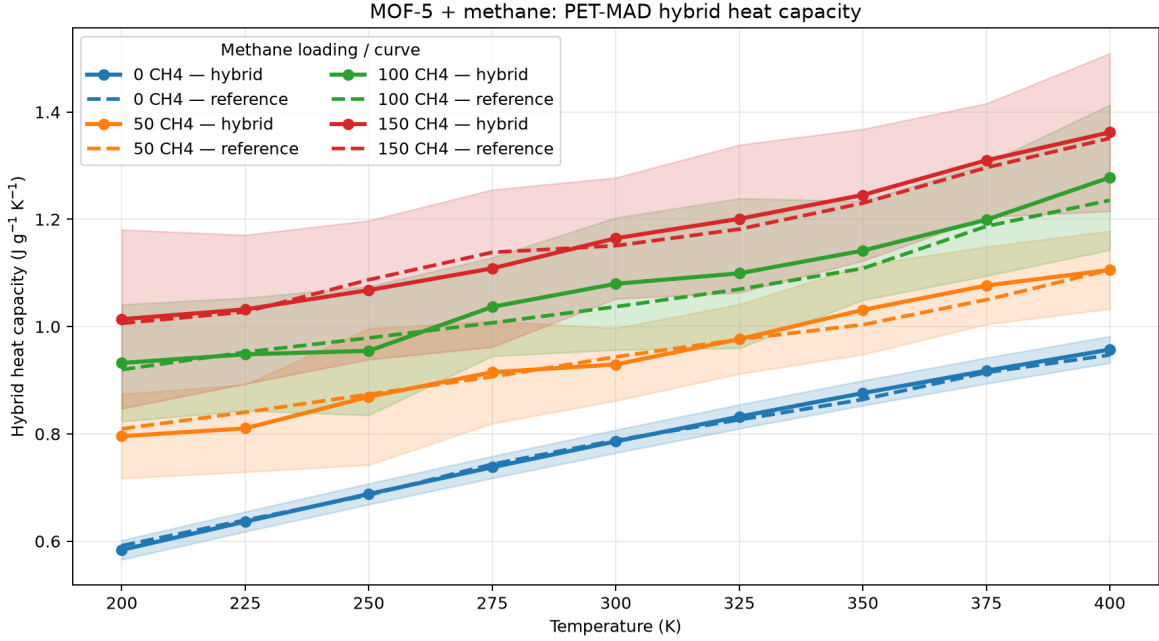
5.6 What determines the uncertainty bands?

Figures 11 and 12 separate the uncertainty contributions. The blue MD sampling error of the enthalpy derivative is small, approximately 0.010 J/g/K at 300 K. The green harmonic LLPR spread is larger and generally increases with loading. The orange classical spread is obtained either by differentiating the CEA member enthalpies [Eq. (65)] or by the Gaussian NPT approximation of Section 2.4.3. The red total uses the standard deviation of the *paired sum* of classical and harmonic member contributions, then combines this with MD sampling error. Thus it need not exceed every individual model contribution.

For example, PET-SOL at 150 molecules and 300 K has classical and harmonic CEA-based model standard deviations of 0.108 and 0.155 J/g/K . Their paired hybrid spread is only 0.103 J/g/K , corresponding to an inferred member correlation of approximately

Table 2: Heat capacities and uncertainty estimates at 300 K, all in J/g/K and normalized by the total mass of each loaded cell. c_{cl} is the central enthalpy derivative, Δc_{har} the retained-mode harmonic correction, and c_{hyb} their sum. u_{CEA} and u_{G} use different paired model spreads around the *same* central value and MD standard error σ_{MD} . The final column is the separate variance-based central hybrid estimate. None of these uncertainties includes the bias from discarded imaginary modes or an estimated spread across minima.

Model	N_{CH_4}	c_{cl}	Δc_{har}	c_{hyb}	σ_{MD}	u_{CEA}	u_{G}	$c_{\text{hyb}}^{\text{var}}$
PET-MAD	50	2.281	-1.351	0.929	0.010	0.068	0.127	0.926
PET-MAD	100	2.762	-1.682	1.080	0.010	0.123	0.169	1.044
PET-MAD	150	3.122	-1.957	1.165	0.010	0.113	0.214	1.202
PET-SOL	50	2.289	-1.363	0.926	0.010	0.087	0.150	0.940
PET-SOL	100	2.762	-1.693	1.069	0.010	0.153	0.225	1.024
PET-SOL	150	3.134	-1.969	1.166	0.010	0.103	0.306	1.230



Exploratory: unstable/near-zero modes discarded; loaded uncertainty includes sampling and LLPR spread.

Figure 9: Supplied PET-MAD loading comparison. Loaded solid curves are central classical enthalpy derivatives plus retained-mode harmonic quantum corrections. Their shading is $\pm u_{\text{CEA}}$, combining MD sampling error with correlated classical-harmonic LLPR spread. Despite the common “hybrid” legend, the zero-loading curve is a quantum-harmonic c_V calculation only, with a harmonic LLPR band; no empty-system MD term was computed. Dashed curves are the supplied reference CSVs, whose provenance is discussed in the text.

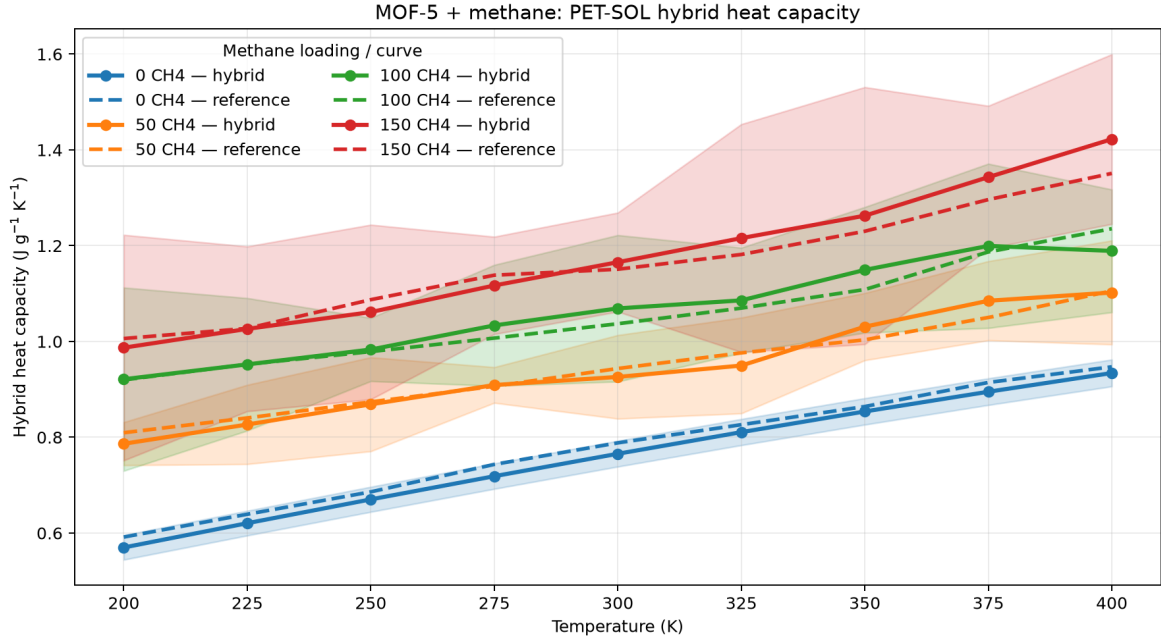
−0.75. Conversely, PET-SOL at 100 molecules has positive correlation (approximately +0.37), increasing the paired spread to 0.153 J/g/K. Summing the separate model variances as if independent would miss both effects. Model spread is not divided by $\sqrt{64}$, because it describes variations of predicted properties across the ensemble rather than the uncertainty of repeated measurements of one fixed mean.

In the refreshed 300 K comparison, Gaussian propagation gives larger combined bands in every model-loading case: for PET-MAD they are 0.127, 0.169, and 0.214 J/g/K, compared with CEA values 0.068, 0.123, and 0.113 J/g/K. For PET-SOL the corresponding Gaussian values are 0.150, 0.225, and 0.306 J/g/K, compared with 0.087, 0.153, and 0.103 J/g/K for CEA. Gaussian classical-harmonic correlations at this temperature are positive in all six cases, approximately 0.43–0.63. The method dependence of both the clas-

sical spread and its covariance is therefore material to the reported uncertainty.

These differences should not be interpreted as proof that the narrower or smoother band is correct. Across all runs and members, the recorded $\text{var}(\beta\Delta U_i)$ ranges from approximately 2.4 to 1.42×10^3 , rather than being much smaller than one as required for a controlled first-order CEA. Direct-reweighting effective sample counts approach one for some members. Consequently, exact reweighting is not a reliable independent validation of CEA for those members, and Gaussian tail behavior also needs checking. The bands are conditional diagnostics of these propagation approximations; their statistical coverage against reference electronic-structure data has not been demonstrated here.

The separate **variance-hybrid** results change the *central* classical estimator as well as its sampling error. Their 300 K hybrid values are listed in the final column



Exploratory: unstable/near-zero modes discarded; loaded uncertainty includes sampling and LLPR spread.

Figure 10: Supplied PET-SOL loading comparison, with the same definitions and exploratory mode treatment as Figure 9. The slight decrease of the 100-molecule central curve at the upper end of the grid is smaller than its uncertainty band and is not evidence of a resolved thermodynamic anomaly. Shading denotes standard uncertainty, not a calibrated confidence interval.

Table 3: Archived central loaded-Hessian diagnostics. All calculations use PET-JAX, float64, three graph hops, and the force-constant row-sum acoustic treatment. “Converged” reproduces the relaxation archive’s flag for its requested tolerance. Imaginary modes have signed frequencies below -1 cm^{-1} ; near-zero modes satisfy $|\tilde{\nu}| \leq 1 \text{ cm}^{-1}$. Retained modes have positive frequencies above the threshold. Signed negative values represent unstable curvature, not physical negative-frequency oscillators.

Model	N_{CH_4}	Converged	F_{max} (eV/Å)	Imaginary	Near zero	Retained	$\tilde{\nu}_{\text{min}}$ (cm^{-1})
PET-MAD	50	no	0.0170	17	5	2000	-53.5
PET-MAD	100	yes	0.0097	114	3	2655	-92.4
PET-MAD	150	no	0.0095	28	3	3491	-66.8
PET-SOL	50	no	0.0054	32	3	1987	-107.7
PET-SOL	100	yes	0.0100	95	3	2674	-138.4
PET-SOL	150	no	0.0049	35	3	3484	-86.5

of Table 2; they differ from the derivative-based values by approximately -0.045 to $+0.064 \text{ J g}^{-1} \text{ K}^{-1}$. Their MD sampling errors are 0.022 – 0.030 J/g/K , larger than those of the enthalpy derivative, and are estimated from the autocorrelation of squared configurational-enthalpy fluctuations. For the central potential this is a fluctuation estimator of the same classical NPT response (Section 2.1.8); Gaussian closure is needed to infer the member-ensemble variances from the central trajectory. Agreement in overall scale is a useful check, but the two estimates share trajectories and harmonic corrections and therefore are not independent measurements.

Taken together, the outputs support a positive loading dependence, a large quantum reduction of the classical heat capacity, and substantial guest mobility. They also identify the immediate requirements for a quantitative prediction: resolve the unstable Hessian spectra,

establish stationarity with longer independent trajectories, sample more quenched structures, and validate propagation under representative LLPR members. Reducing the already small reported MD standard error alone will not resolve these limitations.

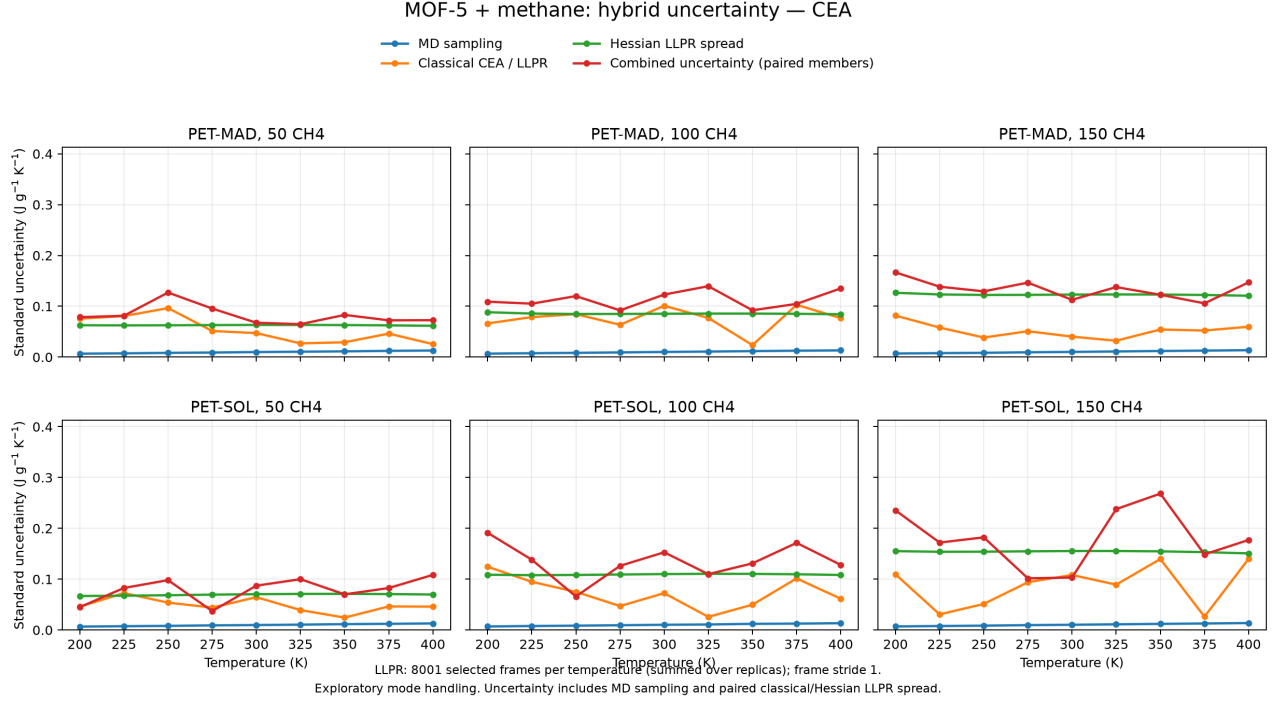


Figure 11: Supplied uncertainty decomposition using refreshed CEA member curves. Every panel uses all 8001 saved production frames at each temperature and 64 LLPR members. The harmonic curve is model spread at one fixed geometry, not an uncertainty across minima. The total preserves classical-harmonic covariance and excludes an unestimated harmonic sampling term. All plotted quantities are standard errors or model standard deviations in J/g/K , not variance fractions.

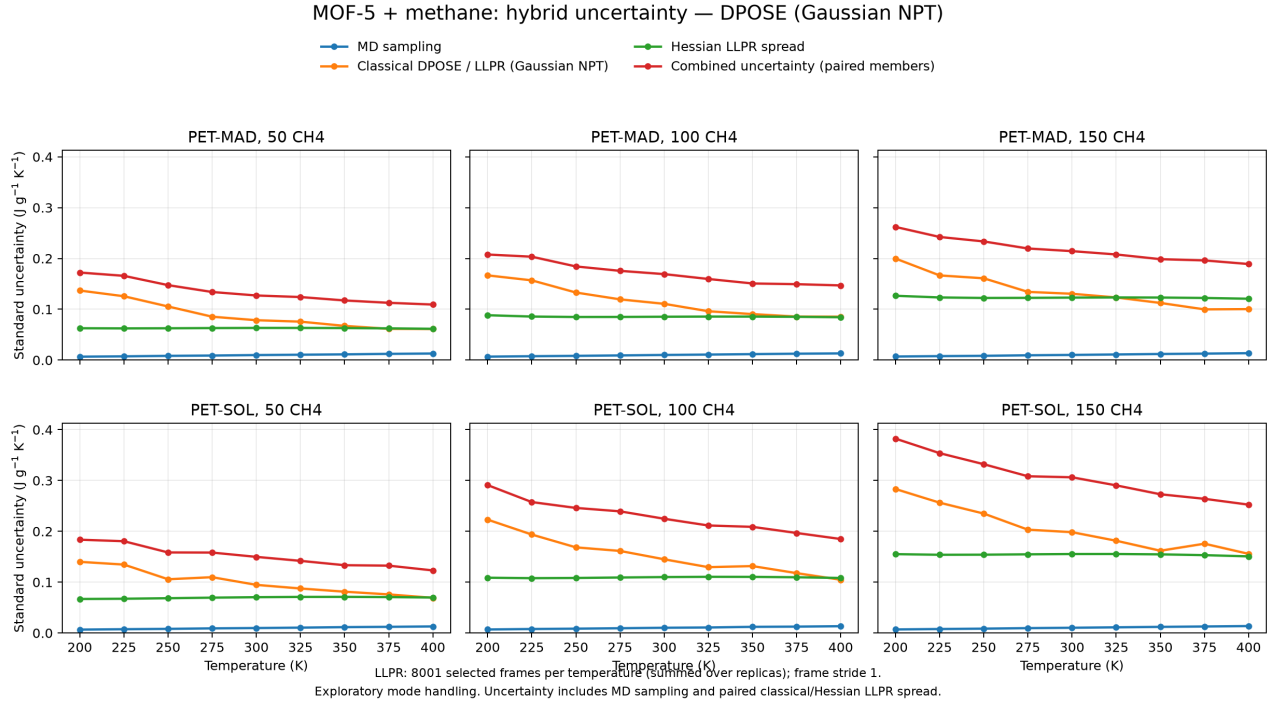


Figure 12: Supplied Gaussian NPT/DPOSE uncertainty decomposition. Central hybrid curves, MD sampling errors, harmonic member corrections, and frame coverage are identical to the CEA comparison; only the classical member spread and its covariance with the harmonic correction change. The smoother orange curves result from a different estimator and do not establish more accurate uncertainty calibration.

6 Recommended production and analysis protocol

6.1 Empty MOF-5 reference

The equilibrated empty-framework structure is used directly; an empty-MOF MD campaign would add cost without supplying the anharmonic loaded-system term needed by Eq. (91).

1. Relax the atomic coordinates with the same PET-MAD checkpoint used for the Hessian. Report maximum residual force and do not relax the cell unless a stress-capable, validated model is adopted.
2. Check stability using signed eigenvalues and inspect low-frequency modes. A large number of unstable modes invalidates the harmonic interpretation.
3. Converge float32 against float64, graph hops, HVP chunking, acoustic sum-rule treatment, and frequency threshold.
4. Converge the conventional-cell result with respect to supercell size or phonon wave-vector sampling. A single periodic cell Hessian samples only the corresponding finite set of modes.
5. Retain the resulting curve as a separately normalized reference and compare its room-temperature value with the approximately 0.7 J/g/K published

reference scale as a contextual check, not a fitting target; PET-MAD replaces QuickFF and its model error remains to be quantified.

6.2 Loaded MOF-5 production and Hessians

For the 100-methane system, the production sequence is:

1. Prepare independent methane placements and velocity seeds for each replica on the enthalpy temperature grid.
2. Run flexible-cell classical NPT only for the loaded system. Determine equilibration from enthalpy, cell, framework, and methane-mobility diagnostics rather than assuming the template duration is sufficient.
3. Average enthalpy over the stationary part of each replica, combine replica uncertainties, and differentiate the temperature dependence.
4. Select representative independent loaded configurations, quench each one to a fixed-cell minimum with the same MLIP, and reject unconverged or unstable minima.
5. Compute one AD Hessian per accepted minimum. Converge precision, graph hops, HVP chunking, acoustic treatment, frequency threshold, and cell size on the quantum-minus-classical correction.
6. Average the loaded corrections, combine them with the classical enthalpy derivative using Eq. (91), and report MD and minimum-to-minimum uncertainties separately as well as jointly.

This calculation applies the hybrid method, rather than attempting an exact reproduction of the reference study. Its success is assessed by internal numerical convergence and by validation of the PET potential for the relevant structures; quantitative accuracy below approximately 200 K remains to be established for the selected MLIP and loading.

6.3 Equilibration and convergence criteria

Production decisions should be based on observables, not a fixed waiting time alone. At minimum, inspect:

- running and block averages of temperature, potential energy, and total energy;
- maximum and distribution of force magnitudes;
- framework bond integrity, pore geometry, and cell consistency;
- methane center-of-mass distributions, host-guest distances, and mean-squared displacements;
- sensitivity to timestep and thermostat friction;

- agreement among independent seeds;
- convergence of the enthalpy derivative with replica count, sampling length, and temperature spacing; and
- convergence of the harmonic correction with independent minima, precision, sparsity depth, cell size, and mode treatment.

The timestep should first be validated in short NVE tests by checking energy drift. Thermostat convergence should then be tested separately, because a thermostat can conceal an integration error while maintaining the desired kinetic temperature.

7 Implementation, outputs, and reproducibility

The production campaign is prepared by `python -m mof_heat_capacity.protocols.loaded` and submitted by `simulation/submit_loaded_md.sh`. These commands create independent loaded structures and LAMMPS/metatomic NPT runs while deliberately rejecting a zero-methane MD campaign. Trajectory diagnostics are submitted with `properties/submit_analysis.sh`.

The command `properties/submit_heat_capacity.sh` takes final structures from selected loaded replicas, relaxes them with `mof_heat_capacity.structures.relax`, and passes the optimized structures to `mof_heat_capacity.analysis.harmonic`. The same command relaxes and evaluates the supplied empty reference directly. Harmonic archives store the run and model identity, optimized structure, temperature grid, frequency array, gravimetric C_V curve, precision, graph hops, HVP chunk size, and rematerialization choice. Finally, `properties/submit_hybrid_analysis.sh` differentiates the loaded NPT enthalpies and writes the classical term, harmonic correction, combined curve, uncertainties, and provenance as NPZ, CSV, and JSON files.

The uncertainty calculations and their stored locations are summarized in Table 4.

The scalar analytical LLPR energy uncertainty is archived but is not used directly as the heat-capacity error. That error is calculated only after CEA has produced a complete enthalpy curve for every persistent ensemble member. The PET-JAX Hessian calculation uses the same persistent LLPR readouts and the hybrid model spread preserves their correlation with the classical CEA term. The Hessians are evaluated at the central-model minima; member-resolved geometry optimization is not included.

Cluster calculations use one node, one task, and one GPU because neither the MD nor Hessian code is MPI or multi-GPU parallel. Independent temperature, replica, or optimized-minimum jobs are the available level of parallelism. Slurm job identifiers and resource use should be retained; elapsed time and peak memory can be recovered with, for example,

```
sacct -j JOB_ID \
  --format=JobID,State,Elapsed,\
  MaxRSS,AllocTRES
```

Wall time and memory requests should be based on representative jobs. Separate output directories and prefixes are required for every loading, temperature, seed, and replica.

A reproducible result should record:

- source structure, cell, atom count, chemical composition, loading, and methane-insertion seed and rejection criteria;
- PET-MAD checkpoint hash, metatomic export, PET-JAX conversion, software environment, device, and numerical precision;
- LLPR covariance and calibration dataset hashes, energy-label definition, regularizer, calibration method, random seed, member count, exported ensemble hash, frame stride, and CEA diagnostics;
- ensemble, target temperature, timestep, thermostat and friction, number of steps, output stride, velocity seed, equilibration interval, production interval, and restart history;
- Hessian source replica, relaxation state, residual forces, graph hops, HVP chunk size, rematerialization, acoustic sum rule, frequency threshold, and temperature grid; and
- independent-replica count, autocorrelation or block length, uncertainty estimator, and all convergence tests.

8 Limitations and interpretation

The results in Section 5 demonstrate the following limitations:

- PET-MAD and QuickFF describe different potential-energy surfaces. Reproducing the paper’s numerical curves is not a project objective, and any agreement or disagreement requires physical interpretation.
- Flexible-cell NPT requires a stress-capable MLIP and a MOF-5-specific stress benchmark; an output capability flag alone is not validation.
- Classical nuclei are used in MD; nuclear quantum effects enter only through the harmonic oscillator formula.
- The random insertion is not an adsorption or Monte Carlo equilibrium distribution.
- A finite number of quenched structures may not represent every thermodynamically important loaded basin. Their spread measures sensitivity but is not automatically a free-energy-weighted average.

Table 4: Uncertainty components, their calculation locations, and their stored outputs.

Quantity	Where it is calculated	Stored result
LLPR energy standard deviation	metatraining LLPR covariance and calibration	<code>energy_uncertainty</code> ; copied as <code>analytical_energy_uncertainty_eV</code> in the per-run archive
CEA member enthalpies and overlap diagnostics	<code>analysis/uncertainty.py</code> , function <code>committee_enthalpy_estimates</code>	per-run <code>model_uncertainty.npz</code>
Classical LLPR C_P standard deviation	<code>analysis/results.py</code> , after member-wise replica averaging and temperature differentiation	<code>model_uncertainty_heat_capacity.npz</code> and <code>.csv</code>
MD sampling standard error	<code>analysis/statistics.py</code> , then replica combination and derivative resampling in <code>analysis/hybrid.py</code>	<code>classical_anharmonic_cp_standard_error</code>
Harmonic sampling standard error	<code>analysis/hybrid.py</code> , across independent optimized minima	<code>harmonic_quantum_correction_standard_error</code>
Hessian LLPR standard deviation	<code>analysis/harmonic.py</code> , after differentiating persistent LLPR readouts at the central minimum	<code>harmonic_quantum_correction_model_standard_deviation</code>
Combined standard uncertainty	quadrature in <code>analysis/hybrid.py</code>	<code>approximate_cp_combined_standard_uncertainty</code>

- The harmonic correction cannot itself represent methane diffusion or barrier crossing; those effects enter only through classical MD.
- Adding a harmonic C_V correction to a classical C_P term assumes that the corresponding pressure-volume quantum correction is small.
- The approximation was quantitatively successful in the reference 100-methane system mainly above about 200 K; transfer to this MLIP, other loadings, and lower temperatures requires validation.
- Finite-size, sparse-Hessian depth, precision, temperature-derivative, model-validity, and independent-replica errors must all be quantified.
- LLPR varies only the final PET readout and cannot measure bias shared by the frozen representation. The present propagation includes paired classical and harmonic readout variations at central-model geometries; it omits member-specific relaxation. Large reweighting diagnostics challenge first-order CEA, and the Gaussian NPT alternative changes the uncertainty substantially.
- Half-trajectory enthalpy diagnostics flag 33 of 66 runs, and one replica per state point cannot establish between-replica uncertainty.
- Every loaded reference spectrum contains imaginary modes; four relaxations are flagged unconverged. Discarding unstable modes does not establish harmonic validity, and one geometry per curve leaves minimum-to-minimum uncertainty unestimated.

These qualifications define the boundary between an efficient, physically motivated approximation and an exact quantum-anharmonic thermodynamic result. They should be carried into every plot, table, and comparison with experiment.

9 Conclusions

The available PET-MAD and PET-SOL calculations apply the hybrid approximation to three methane loadings. Both give increasing room-temperature heat capacity with loading, from about 0.93 to 1.16 J/g/K, after a harmonic quantum correction removes roughly 59–63% of the classical contribution. The trajectory diagnostics show substantial methane mobility and a loading-dependent cell response, supporting the inclusion of anharmonic MD. The calculated trends are broadly consistent with the digitized reference curves, without establishing equivalent accuracy for the different potentials.

The current heat-capacity curves remain exploratory. Temperature control is good, but production-half discrepancies, a single replica per state point, imaginary Hessian modes, and incomplete relaxations prevent a claim of converged thermodynamics. Classical and harmonic LLPR uncertainties must be combined with their covariance; changing from CEA to Gaussian NPT propagation substantially changes the resulting bands. Stable and converged reference spectra, independent trajectory and minimum sampling, and validation of member propagation are required before these numerical values can be treated as quantitative predictions. The separately calculated empty-framework harmonic curve remains a reference and does not replace the loaded correction.

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